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PERFORMANCE SPECIFICATION

TEST EQUIPMENT FOR USE WITH ELECTRICAL AND ELECTRONIC EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for, but is not limited to, test equipment (see 6.6.28) used in testing and calibrating electrical and electronic equipment. The test equipment may be of commercial design and includes general purpose, special purpose, peculiar, console mounted, automatic test equipment (ATE) (see 6.6.3), and calibration standards. This specification should also be used for built-in test equipment (BITE) (see 6.6.6) when the requirements for a built-in test (BIT) (see 6.6.5) are not included in the system specification. Detailed requirements for particular test equipment should be as specified (see 6.2).

1.2 Classification. Test equipment covered by this specification is categorized by performance characteristics required for an intended location of use and the most severe prevailing environmental conditions of such locations. Test equipment are of the following classes as specified (see 6.2), and the performance characteristics required for each class are summarized in table II:

a. Class 1 - Equipment is designed for hostile operational environments where the environmental conditions are extremes of world climatic variation. Examples include test equipment for use in the vicinity of aircraft, in tracked and wheeled vehicles, or for typical above-deck applications.

b. Class 2 - Equipment is designed for rugged (see 6.6.22) operational environments where the environmental conditions are routinely encountered in an unprotected, uncontrolled climate. Test equipment for use as a portable (see 6.6.20) instrument in more than one location that has a substantial degree of environmental protection may be considered as class 2.

c. Class 3 - Equipment is designed for operational environments where conditions are routinely encountered in a diverse, controlled climate. Examples include test equipment for bench-top (see 6.6.4), rack-mount, or occasional portable use in diverse and environmentally controlled locations.

d. Class 4 - Equipment is designed for benign operational environments where the environmental conditions are controlled and protected. Examples include test equipment for use in a fully protected and environmentally controlled service area such as a laboratory.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9310-AQ-SAF-010 - Technical Manual for Batteries, Navy Lithium Safety Program Responsibilities and Procedures

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ELECTRONIC COMPONENTS INDUSTRY ASSOCIATION (ECIA)

EIA/ECA-310 - Cabinets, Racks, Panels, and Associated Equipment

(Copies of this document are available online at https://global.ihs.com.)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE 315 - Graphic Symbols for Electrical and Electronic Diagrams (Including Reference Designation Letters

IEEE 488.1 - Higher Performance Protocol for the Standard Digital Interface for Programmable Instrumentation – Part 1: General

IEEE 488.2 - Standard Digital Interface for Programmable Instrumentation – Part 2: Codes, Formats, Protocols and Common Commands

IEEE 802.3 - IEEE Standard for Ethernet

(Copies of these documents are available online at www.ieee.org.)

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 61000-4-2 - Electromagnetic Compatibility (EMC) – Part 4-2: Testing and Measurement Techniques – Electrostatic Discharge Immunity Test

IEC 61000-4-3 - Electromagnetic Compatibility (EMC) – Part 4-3: Testing and Measuring Techniques – Radiated, Radio-Frequency, Electromagnetic Field Immunity Test

IEC 61000-4-4 - Electromagnetic Compatibility (EMC) – Part 4-4: Testing and Measurement Techniques – Electrical Fast Transient/Burst Immunity Test

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IEC 61000-4-6 - Electromagnetic Compatibility (EMC) – Part 4-6: Testing and Measurement Techniques – Immunity to Conducted Disturbances, Induced by Radio-Frequency Fields

IEC 61010-1 - Safety Requirements for Electrical Equipment for Measurement, Control, and Laboratory Use – Part 1: General Requirements

IEC 61010-031 - Safety Requirements for Electrical Equipment for Measurement, Control and Laboratory Use – Part 031: Safety Requirements for Hand-Held and Hand-Manipulated Probe Assemblies for Electrical Test and Measurement

IEC CISPR 11 - Industrial, Scientific and Medical Equipment – Radio-Frequency Disturbance Characteristics – Limits and Methods of Measurement

(Copies of these documents are available online at www.iec.ch.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 7779 - Acoustics – Measurement of Airborne Noise Emitted by Information Technology and Telecommunications Equipment

(Copies of this document are available online at www.iso.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 60950-1 - Information Technology Equipment – Safety – Part 1: General Requirements

(Copies of this document are available online at www.comm-2000.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Equipment sample requirements.

3.1.1 Bid sample. When specified (see 6.2), a bid sample shall be subjected to bid sample inspection (see 6.3) in accordance with 4.2.

3.1.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Equipment components. Parts, materials, and processes shall be as specified herein and shall permit the test equipment to meet all applicable requirements of its class.

3.2.1 Finish. Unless otherwise specified (see 6.2), the equipment shall be the finish normally provided by the manufacturer.

3.2.2 Color. Unless otherwise specified (see 6.2), the color shall be the color normally provided by the manufacturer.

3.2.3 Interchangeability. A model number shall be designated for the units (see 6.6.28) provided. All units and the replaceable assemblies, subassemblies, and parts comprising the unit shall be interchangeable on a noninterference basis with all such designated models provided.

3.2.4 Restricted materials. Test equipment furnished to the Navy that uses material restricted for Navy use (see 3.2.4.1) or that exceeds the size limitations (see 3.7.1) of this specification requires NAVSEA approval prior to procurement.

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3.2.4.1 Material restricted for Navy use. Unless otherwise specified (see 6.2), the following materials and parts shall not be used in equipment for the Navy:

a. Mercury, including mercury batteries

b. Radioactive material

c. Lithium electrochemical cells

d. Helium-3

e. Cadmium, except batteries

3.2.4.2 Other restricted materials and gases. Test equipment shall not contain materials that promote the premature failure of the equipment or that present a health hazard under normal use or storage conditions as specified herein.

3.2.4.2.1 Organic materials. Test equipment shall not contain materials prone to organic decay or that act as a nutrient for the growth of multicelled organisms.

3.2.4.2.2 Flammable materials. Test equipment shall not contain materials that ignite into flames at operating or storage temperatures.

3.2.4.2.3 Explosive materials. Test equipment shall not contain materials that explode at operating or storage temperatures.

3.2.4.2.4 Carcinogens. Certain chemicals have been identified by the Occupational Safety and Health Administration (OSHA) in CFR Title 29 Part 1910 as cancer-producing substances (carcinogens). Electrical test equipment shall not contain carcinogenic materials.

3.2.4.2.5 Helium. Helium shall not be used as pressurizing gas in sealed units containing electron tubes. When necessary to use helium for leak detection purposes, exposure shall be limited to the time necessary for the test followed by thorough purging. Helium-3 shall not be used due to limited availability.

3.2.4.3 Combination of materials. The equipment shall not contain any combination of materials that cause deterioration of any material contained in the equipment due to the effects of outgassing.

3.2.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction.

3.3.1 Operator adjustments. The equipment shall include built-in adjustments or compensating devices for any function or parameter that would exceed the specified tolerance within the calibration interval defined in 3.13. Under all specified service conditions, the variation of any adjustment or control shall produce the desired control action to bring the equipment within tolerance.

3.3.2 Operator controls. Operating controls and indicators shall be readily accessible to the operator from the front of the equipment. The readings presented on the controls and indicators shall be available to the operator without conversion or calculation.

3.3.3 Electric connectors. Provision shall be made to ensure that connectors can only be mated with the appropriate counterparts. Where design considerations require close proximity of connectors of similar configuration, the mating connectors shall be suitably coded or marked. The live or hot side of unmated connectors shall be protected against shorting.

3.3.4 Internal wiring. Internal wiring practices shall be such that the wiring shall not be damaged during normal maintenance, filter service, and calibration.

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3.3.5 Internal cooling. The equipment shall be capable of operating without interruption for at least 24 hours (or maximum battery life) at the maximum operating temperature for the applicable class. Cooling devices used in the equipment shall be an integral part of the equipment and shall not degrade any aspect of performance or safety.

3.3.6 Front panel display indicators. Front panel display indicators designed to present change-of-state information to the operator shall be distinguishable at a distance of 1 meter (m) in a maximum ambient light level of 100 foot-candles (fc) and a minimum light level of 1 fc. All display indicators shall be discernible at viewing angles from 0 to 45 degrees.

3.3.7 Self-test capability. The equipment shall be able to provide automatic diagnostic self-test (see 6.6.25) information. At a minimum, the information provided shall indicate whether the equipment is functional or identify malfunctions. Passive devices (see 6.6.18) are not required to have a self-test function.

3.4 Cybersecurity. Cybersecurity requirements shall be as specified (see 6.2).

3.5 Digital control interface. When specified (see 6.2), a digital control interface shall be provided. The measurement information to be transferred by the digital control interface shall be as specified (see 6.2). The digital control interface shall have the programmable capability to control each device function including all ranges, modes, and levels. Primary power control and other non-measurement functions need not be controllable through the digital control interface.

3.5.1 IEEE 488 interface requirements. If an IEEE 488 digital control interface is specified (see 6.2), it shall comply with IEEE 488.1 and with the specified portions of IEEE 488.2. The interface functions and subset identifications specified below shall be considered for determining equipment requirements.

Interface function	Subset identification
Source handshake	SH1
Acceptor handshake	AH1
Talker	T1 or TE1
Listener	L1 or LE1
Service request	SR1
Remote local	RL1
Parallel poll	PP0
Device clear	DC1
Device trigger	DT1
Controller	C0
Configuration	CF

3.5.2 IEEE 802.3 ethernet interface requirements. If an IEEE 802.3 digital control interface is specified (see 6.2), it shall comply with the 10/100/1000 Base-T communications sections of IEEE 802.3.

3.5.3 USB interface requirements. If a USB digital control interface is specified (see 6.2), it shall conform to the standards specified by the Universal Serial Bus Implementers Forum (USB-IF).

3.6 Electrical power sources and connections.

3.6.1 Electrical power source. The equipment shall operate from the input power (see 6.6.12) conditions specified herein. When multiple power source requirements are invoked, the equipment shall operate with each required source without changing manually operated devices for compensation, such as tapped transformers (variable setting transformers) and other similar devices. When multiple power source requirements are invoked, all power source settings shall be automatically selected or externally selectable.

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3.6.1.1 Power consumption. The maximum power for operation shall be as specified (see 6.2).

3.6.1.2 Nominal and alternate power source. Equipment that does not have a battery as the primary power source shall operate at 120 volts root mean square (Vrms), single-phase, 50 hertz (Hz) and 60 Hz source (see table I). Alternate power source requirements, which may be required for equipment, are specified as follows:

a. 50 Hz, 60 Hz, and 400 Hz single-phase 120 Vrms.

b. 50 Hz, 60 Hz, and 400 Hz single-phase with voltage selectable from a 100/120/220/240 Vrms range.

c. 50 Hz, 60 Hz, and 400 Hz single-phase 120/240 Vrms (see 3.6.1.2.4).

d. Direct current (DC) internal power source (see 3.6.1.2.5).

e. DC external power source (see 3.6.1.2.6).

Equipment designed to operate from both external and internal power sources shall conform to specified performance and accuracy requirements when operated from the external source with the internal battery removed. There shall be an indication of the selected power source.

TABLE I. Power source.

Voltage (Vrms)			
	Steady-state	**Transient-state**	**Interruption**
Nominal	108 to 132	84 to 108 132 to 156	0 to 84
Alternate (see 3.6.1.2.4)	216 to 264	168 to 216 264 to 312	0 to 168
Frequency (Hz)			
	Steady-state	**Transient-state**	**Interruption**
Nominal	47.5 to 52.5	45 to 47.5 52.5 to 57	0 to 45
	57 to 63	54 to 57 63 to 66	0 to 54
Alternate (see 3.6.1.2.4)	380 to 420	360 to 380 420 to 440	0 to 360

3.6.1.2.1 Steady-state conditions. The performance and accuracy of the equipment shall not be adversely affected when operating under any combination of the following conditions:

a. The steady-state voltage variation (see 6.6.12.d) shall be ±10 percent of the nominal power source voltage as specified in table I.

b. The steady-state frequency variation (see 6.6.12.d) shall be ±5 percent of the nominal power source frequency as specified in table I.

c. The deviations in waveform characteristics shall be as follows:

(1) Waveform deviation factor of ±10 percent or less (see 6.6.12.a)

(2) Total harmonics of ±10 percent or less (see 6.6.12.b(1))

(3) Individual harmonics of ±5 percent or less (see 6.6.12.b(2))

(4) Crest factor of 1.27 to 1.56 (see 6.6.12.c)

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3.6.1.2.2 Transient-state conditions. For transients that do not exceed the nominal steady-state band specified in table I, the transient recovery time (see 6.6.12.f) of the equipment shall not exceed 500 milliseconds (ms). Operation of the equipment during voltage and frequency transients that exceed the allowable nominal steady-state bands specified in table I shall be in accordance with the following:

a. The maximum transient amplitude (see 6.6.12.e) of voltage transients within the range of ±10 percent to ±30 percent shall not cause the operation of the equipment to be impaired for longer than 30 seconds following the transient and shall not cause loss of information stored in volatile memory. Testing shall be in accordance with 4.5.5.6.3.

b. The maximum transient amplitude (see 6.6.12.e) of frequency transients within the range of ±5 percent to ±10 percent shall not cause the operation of the equipment to be impaired for longer than 30 seconds following the transient and shall not cause loss of information stored in volatile memory. Testing shall be in accordance with 4.5.5.6.4.

The equipment performance may be impaired during the transient. Transient values are derived from and referenced to the nominal values specified for voltage and frequency.

3.6.1.2.3 Interruption of power source. Voltage or frequency interruptions shall not cause damage to the equipment. A power source interruption is when the voltage falls in the range of 0 to 70 percent of nominal or frequency falls in the range of 0 to 90 percent of nominal. Equipment performance is not required during interruption of the power source. After power interruptions of any duration, the equipment shall automatically resume operation at either the last operating condition before interruption or a default power-up condition.

3.6.1.2.4 Alternating single-phase source. When the equipment is designed to operate from a 240-Vrms nominal or 400-Hz nominal alternating single-phase source, the tolerance and times of the steady-state, transient-state, and interruption of power source requirements shall apply to the alternate power source.

3.6.1.2.5 DC internal power source. When specified (see 6.2), the equipment shall be designed to operate from self-contained batteries.

3.6.1.2.5.1 Battery restrictions. Battery restrictions shall apply as follows:

a. Standard batteries. Unless otherwise specified (see 6.2), maximum use shall be made of standard, nomenclatured battery-power sources.

b. Mercury batteries. Unless otherwise specified (see 6.2), the use of mercury batteries is restricted and these shall not be used in equipment. If a restricted type of battery is specified, the Use and Installation Manual (see 6.6.32) and the Maintenance and Service Manual (see 6.6.15) shall reference substitute battery types and all necessary modifications to use the substitute batteries. The equipment shall incorporate safety precautions of UL 60950-1 and a label containing the following caution and battery information shall be affixed to the outside of the unit:

THIS UNIT CONTAINS MERCURY BATTERIES Battery Manufacturer: Battery Part Number:

c. Lithium batteries. S9310-AQ-SAF-010 establishes the safety guidelines for the usage of lithium batteries. Unless otherwise specified (see 6.2), the use of lithium batteries is prohibited without prior authorization. S9310-AQ-SAF-010 promulgates the exception requirements for approving the usage of lithium batteries If equipment qualifies for one or both of the documented exceptions, then the required information shall be as specified (see 6.2) at the time of submission of proposals. If equipment does not qualify for an exception, then a request for approval for the use of lithium batteries shall be as specified (see 6.2). Approval shall apply only to the specific model proposed.

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The lithium ion rechargeable cell manufacturer, part number, configuration (number of cells in series), nominal voltage, and rated capacity shall be as specified (see 6.2) at the time of submission of proposals. The equipment shall incorporate safety precautions of UL 60950-1 and a label containing the following caution and battery information shall be affixed to the outside of the unit:

THIS UNIT CONTAINS LITHIUM BATTERIES Battery Manufacturer: Battery Part Number:

3.6.1.2.5.2 Battery information. If batteries are used in the equipment, instructions shall be provided that describe the procedure for battery replacement, including the part number of the battery and description of the replacement process. If lithium batteries are used, proper disposal procedures shall be contained in the Use and Installation Manual and the Maintenance and Service Manual.

3.6.1.2.5.3 Battery state indicator. A battery state indicator shall be provided to indicate when the capacity of batteries used for primary power is low. The approximate remaining operating time for the low battery state shall be specified in the Use and Installation Manual.

3.6.1.2.5.4 External battery pack. When the equipment is required to operate from an external portable battery pack, the battery pack shall be as specified (see 6.2).

3.6.1.2.6 DC external power source. Equipment operation from an external DC source shall be as specified (see 6.2).

3.6.2 Electrical power connections. The requirements specified herein shall apply to equipment rated at 10 amperes (A) rms or less. Power connectors and connections for equipment rated above 10A rms shall be as specified (see 6.2). When single-phase operation is specified, the equipment shall operate with a ground at the primary power source whether from an ashore site that has “grounded circuitry” or afloat where the ship has “ungrounded circuitry.

3.6.2.1 Multiphase equipment. Except as specified for leakage current (see 3.11), equipment requiring multiphase power shall be as specified (see 6.2).

3.6.2.2 Input power cable. Unless otherwise specified (see 6.2), the appropriate input power cable shall be a detachable power cable normally provided by the manufacturer.

3.6.2.3 Input power switch. For bench top equipment, the operator shall be able to energize the equipment from the front panel; a standby (see 6.6.27) capability may be incorporated to activate and deactivate the equipment. If such a capability is included, warnings shall be placed in the Use and Installation Manual and on the equipment to warn the operator of possible shock hazards during servicing due to partially activated power supplies.

3.6.2.4 Power selection switch. When the configuration of the equipment requires a power selection switch, the equipment shall include options to operate at 120 Vrms or 240 Vrms. A safety mechanism shall be incorporated to prevent accidental switching between voltage selections.

3.6.2.5 Excessive current drain protection. Automated safety mechanisms shall be incorporated to automatically disconnect the equipment in the event of current being drawn that is in excess of the equipment rating. The mechanism shall be in the front for class 1 equipment or in an exterior panel for class 2, class 3, and class 4 equipment.

3.6.2.6 Indicators. The front panel shall include visual indicators that indicate when the equipment is energized.

3.6.2.7 Battery operating and recharge time. When battery operation is required, the operating and recharge time shall be as specified (see 6.2).

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3.6.2.8 Battery charger. When specified (see 6.2), the equipment shall be provided with an internal battery charger capable of charging the internal batteries when the equipment is connected to the specified alternating current (AC) power source. Unless otherwise specified (see 6.2), the recharging time to 90 percent capacity provided by the battery charger shall be less than or equal to 75 percent of the operating time provided by a fully charged battery. The charging circuit shall not load the internal battery when the equipment is disconnected from the AC power source. If mechanical or electrical disconnection of the charging circuit is required by the design, disconnection shall be allowed by means other than manual switching. When rechargeable batteries and a charger are required, an automatic protective device shall be included to prevent overcharging the batteries.

3.6.2.9 Battery compartment. When specified (see 6.2), the equipment powered from internal batteries shall include an externally accessible, covered battery compartment. The compartment shall only provide access to the battery and features required for normal operator service, such as fuse replacement, but shall not allow for user alteration of calibration. The battery compartment shall provide protection from external sources that could damage the battery, protect the equipment from deteriorated batteries, and provide access to clean the compartment. The battery polarity, voltage, and type shall be marked in a prominent place on or adjacent to the battery compartment.

3.7 Enclosure physical characteristics. Unless otherwise specified (see 6.2), a commercial-off-the-shelf (COTS) enclosure (see 6.6.9), normally provided by the manufacturer, that meets the requirements herein shall be acceptable.

3.7.1 Dimensions. Dimensions shall be as specified (see 6.2). The required dimensions for the height, width, and depth may be specified for both maximum and partial dimensions (see 6.6.8). For Navy use in shipboard applications, the maximum dimensions shall not exceed a size capable of passage through hatches 508 millimeters (mm) (20 inches) wide by 965 mm (38 inches) high, reduced further by round corners on a 254-mm (10-inch) radius and through circular openings 635 mm (25 inches) in diameter.

3.7.2 Weight. The maximum weight shall be as specified (see 6.2). When a mainframe (see 6.6.28.1) and plug-in (see 6.6.28.2) configuration is supplied, the combined weight shall not exceed the maximum weight allowed. If the weight exceeds 20 kilograms (kg) (44 pounds), a label containing the following cautionary note and weight information shall be affixed to the outside of the unit:

TWO-PERSON LIFT THIS UNIT EXCEEDS 20 KG (44 LBS) AND REQUIRES TWO PERSONS TO MOVE

3.7.3 Mechanical stability. Equipment shall be designed to preclude tipping during normal handling and operation. Testing shall be in accordance with 4.5.7.4. This requirement is not applicable to test equipment smaller than 2000 cubic centimeters (cc) with a weight less than 2.5 kg and excludes equipment that has a depth of 4.5 or more times the height or that has a height of less than 100 mm.

3.7.3.1 Attitude. Equipment shall be mechanically stable when positioned in its normal attitude on a bench for horizontal viewing of the front panel and when placed in its alternate attitude on the floor or deck for vertical viewing of the front panel. The size and weight limitations of 3.7 apply to this requirement.

3.7.4 Enclosure requirements. The degree of protection provided by the enclosure shall allow the equipment to pass the environmental tests applicable to the specified class.

3.7.4.1 Cover. When specified (see 6.2), the enclosure shall have a cover.

3.7.4.2 Latches. When latches are incorporated in the design of the equipment, they shall incorporate a quick-opening mechanism.

3.7.4.3 Stacking provisions. Equipment enclosures shall have a geometric configuration that permits stacking without harm to the enclosure or its contents. The maximum stacking height (expressed as a quantity of equipment in excess of one) shall be as specified (see 6.2).

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3.7.4.4 Component protection. Panels shall be designed to conform to the bench handling requirement of 3.9.5.3.

3.7.4.5 Handles. Unless otherwise specified (see 6.2), one or more handles shall be provided. The number and location of handles shall be such that the load distribution per handle shall not exceed 20 kg. Handles shall be located above the equipment center of gravity to ensure carrying stability. Handles shall permit compliance with the enclosure attitude provisions specified in 3.7.3.1.

3.7.4.6 Accessory stowage. When specified (see 6.2), the enclosure shall be capable of stowing all accessories (see 6.6.1). Stowage shall not adversely affect the carrying stability of the equipment and shall be contained in a manner that prevents damage to the accessories or equipment during transit. Stowed items shall be available without disturbing enclosure seams.

3.7.4.7 Corners. All corners shall be designed to preclude injury to personnel or damage to material. All corners shall be reinforced to protect the equipment from damage during the specified environmental conditions.

3.7.4.8 Connections and controls. Unless otherwise specified (see 6.2), all external electrical connections (except power connections) and operating controls shall be on the instrument panel. Electrical power connections shall be on the rear.

3.7.4.9 Chassis fasteners. Unless otherwise specified (see 6.2), the chassis shall be securely fastened within the enclosure conforming to the maintainability requirements of 3.15.3.

3.7.4.10 Pressure equalizing valve. Unless otherwise specified for class 1 equipment (see 6.2), a mechanism shall be provided for equalizing air pressure within the enclosure when the enclosure is sealed. The mechanism provided shall not protrude beyond the enclosure.

3.7.4.11 Console cabinet enclosures. When specified (see 6.2), the equipment shall be fitted with a console cabinet. The console cabinet enclosure shall provide protection for the contained equipment from the specified environmental conditions. Access to the interior of the enclosure shall be provided. With the equipment installed in the console cabinet, the console cabinet enclosure shall be designed so that the console cabinet can be lifted and transported by forklift vehicles when the console cabinet is in its normal upright position. Additional requirements for transportability shall be as specified (see 6.2).

3.7.4.12 Console cabinet work surfaces. When specified (see 6.2), equipment fitted with a console cabinet shall have work surfaces for personnel provided. Personnel work surfaces shall be constructed of nonconducting material. Any work surface that projects from the console cabinet shall not be damaged during or interfere with transport or shipping.

3.7.4.13 Rack-mounting requirements.

3.7.4.13.1 Rack-mountable enclosures. Enclosures that are required to be rack-mounted and enclosures required to accommodate rack-mountable equipment shall be in accordance with EIA/ECA-310 for a 482.6-mm (19-inch) standard panel width.

3.7.4.13.2 Convertible/rack-mountable equipment. When specified (see 6.2), the equipment shall be capable of being rack-mounted. All convertible/rack-mountable equipment shall conform to the requirements of its class when configured for rack-mounting use in accordance with 3.7.4.13.3.

3.7.4.13.3 Rack-mount conversion. Conversion shall be accomplished by use of a kit that converts the enclosure for rack-mounting but will not prevent return to the original configuration. The rack-mount conversion kit shall include all hardware required to configure the equipment for rack-mounting. When the height of the front panel is not a multiple of the preferred panel height increment or when the width is less than rack-width, the conversion kit shall include filler panels to provide the preferred height and width in accordance with 3.7.4.13.1. No part of the equipment shall protrude above or below the front panel when the enclosure is rack-mounted.

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3.8 Transit case. When specified (see 6.2), all handheld (see 6.6.11) equipment and other types of equipment shall be fitted with a transit case. The transit case shall provide protection for the contained test equipment from the specified environmental conditions. The transit case is not an integral part of the test equipment. The transit case need not provide for operational capability of the equipment when enclosed. The physical characteristics of the transit case shall be as specified herein.

3.8.1 Transit case cover. The transit case shall have a cover that can be closed and fastened. Fasteners shall allow for quick opening. If the cover is removable, fasteners shall not interfere with cover removal.

3.8.2 Transit case stacking provisions. The transit case shall have a geometric configuration that permits stacking without harm to the enclosure or its contents. The maximum stacking height (expressed as a quantity of equipment in excess of one) shall be as specified (see 6.2).

3.8.3 Transit case handles. One or more handles shall be provided. The number and location of handles shall be such that the load distribution per handle shall not exceed 20 kg (44 pounds). Handles shall be located above the equipment center of gravity to ensure carrying stability. Handles shall permit compliance with the enclosure attitude provisions specified in 3.7.3.1.

3.8.3.1 Hard transit case handles. A transit case shall include hinged handles with sufficient internal clearance to allow a block 44 mm (1.75 inches) by 106 mm (4.17 inches) in cross-section with edges rounded to a 25-mm (1-inch) radius to pass through them. The grip portion of the handles shall be of a nonmetallic material at least 89 mm (3.5 inches) in length with a 19-mm (¾-inch) diameter or other approved cross-section that is shaped to fit the hand comfortably. Handles shall stop open at 90 degrees and have a retaining mechanism to keep them in the closed and 90-degree position. Handles shall be recessed or protected. A handle located on the top of the enclosure shall be recessed to facilitate stacking.

3.8.3.2 Soft transit case handles. When a soft transit case is specified (see 6.2), one or more handles shall be provided. A shoulder strap may be provided in addition to the required handles. Handles shall be attached to the case in a manner to ensure serviceability over the life of the equipment under normal conditions of use.

3.8.4 Transit case accessory stowage. The transit case shall allow for the stowage of accessories, operating instructions, removable power cables, signal leads, and designated spares. Stowage shall not adversely affect the carrying stability of the equipment and shall be contained in a manner that prevents damage to the accessories or equipment during transit.

3.8.5 Transit case corners. All hard-edged corners shall be designed to preclude injury to personnel or damage to material. All corners shall be adequately reinforced to protect the instrument from damage during the environmental conditions specified.

3.8.6 Watertight and airtight transit case pressure equalizing valve. Watertight and airtight transit cases shall have a mechanism for equalizing air pressure within the enclosure when the enclosure is sealed. The mechanism shall not protrude beyond the transit case enclosure.

3.9 Environmental requirements. Environmental performance requirements shall be as specified herein and in table II.

3.9.1 Environmental conditions. Equipment shall conform to the specified performance requirements (see 6.6.19) when subjected to the environmental conditions indicated in this specification. Combinations of environments beyond those covered by the specified test procedures (see 4.5 and 6.6.30) shall be as specified (see 6.2). Unless otherwise specified (see 6.2), equipment shall conform to the specified performance requirements when subjected to a 30-minute warm-up period preceded by a 2-hour not operating temperature stabilization period or when tested in accordance with the applicable test method (see section 4).

3.9.2 Temperature and humidity (T/H). The temperature ranges and humidity limits for both operating and not operating conditions shall be as specified herein. A relative humidity (RH) of 95 percent (with the applicable tolerance) shall be achieved in a non-condensing condition.

3.9.2.1 Temperature, not operating. Equipment shall conform to the specified performance and accuracy requirements after being stored at the temperatures specified herein for the applicable class.

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3.9.2.1.1 Class 1 and 2. Class 1 and 2 equipment shall be subjected to not operating temperatures ranging from -51 to 71 °C.

3.9.2.1.2 Class 3 and 4. Class 3 and 4 equipment shall be subjected to not operating temperatures ranging from -40 to 71 °C.

TABLE II. Environmental requirements by class.

Environmental conditions	Class [1/]			
	1	2	3	4
Temperature, not operating (°C) (3.9.2.1 and 4.5.5.1)	-51 to 71	-51 to 71	-40 to 71	-40 to 71
Temperature, operating (°C) (3.9.2.2 and 4.5.5.1)	-40 to 55 [2/]	-10 to 55	0 to 50	10 to 40
RH (percent) (3.9.2.3 and 4.5.5.1) [3/]	5 to 95±5 [4/, 5/]	5 to 95±5 [6/, 7/]	5 to 95±5 [6/, 7/]	5 to 95±5 [6/, 7/]
Altitude, not operating (3.9.3 and 4.5.5.2)	4600 meters	4600 meters	4600 meters	4600 meters
Altitude, operating (3.9.3 and 4.5.5.2) [8/]	4600 meters	4600 meters	4600 meters	3000 meters
Vibration limits (3.9.4.1 and 4.5.5.3.3)	Random 10-2000 Hz	Random 10-500 Hz	Random 5-500 Hz	Random 5-500 Hz
Loose cargo bounce (3.9.4.3 and 4.5.5.3.3)	Yes	Yes	[9/]	[9/]
Functional shock (3.9.5.1 and 4.5.5.4.1)	30g [10/]	30g [10/]	30g [10/]	30g [10/]
Transit drop (3.9.5.2 and 4.5.5.4.2)	Yes	[11/]	[11/]	[11/]
Bench handling (3.9.5.3 and 4.5.5.4.3)	Yes	Yes	Yes	Yes
High-impact shock (3.9.5.4 and 4.5.5.4.4)	[9/]	[9/]	[9/]	[9/]
Watertight (3.9.6.1 and 4.5.5.5.1)	Yes	[9/]	[9/]	[9/]
Splash-proof (3.9.6.2 and 4.5.5.5.2)	Yes	[9/]	[9/]	[9/]
Drip-proof (3.9.6.3 and 4.5.5.5.3)	N/A	[9/]	[9/]	[9/]
Fungus resistance (3.9.7 and 4.5.6.1) [9/]	Yes	Yes	Yes	Yes
Salt exposure, enclosure (3.9.8.1 and 4.5.6.2.1) [9/]	48 hours	48 hours	48 hours	48 hours
Salt exposure, structural parts (3.9.8.2 and 4.5.6.2.2)	48 hours	48 hours [9/]	48 hours [9/]	48 hours [9/]
Explosive atmosphere (3.9.9 and 4.5.6.3)	Yes	[9/]	[9/]	N/A
Dust resistance (3.9.10 and 4.5.6.4)	Yes	[9/]	[9/]	N/A
Solar radiation (3.9.12 and 4.5.6.8)	[9/]	[9/]	N/A	N/A

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TABLE II. Environmental requirements by class – Continued.

NOTES:

1/ See 3.9 for complete environmental requirements.

2/ 20-minute operation at 71 °C.

3/ RH not controlled below 10 °C.

4/ 85±5% RH above 30 °C.

5/ 60±5% RH above 40 °C.

6/ 75±5% RH above 30 °C.

7/ 45±5% RH above 40 °C.

8/ Allowances shall be made for derating of high voltage accuracy to achieve required altitude denoted by a warning label viewable by the operator.

9/ When specified (see 6.2).

10/ Half-sine shock pulse (see 3.9.5.1).

11/ See 3.9.5.2, 4.5.5.4.2, and table XIII.

3.9.2.2 Temperature operating. Equipment shall conform to the specified performance and accuracy requirements while operating at the temperatures specified herein for the applicable class.

3.9.2.2.1 Class 1. Class 1 equipment shall be subjected to operating temperatures ranging from -40 to 55 °C. Class 1 test equipment shall operate for 20 minutes at 71 °C.

3.9.2.2.2 Class 2. Class 2 equipment shall be subjected to operating temperatures ranging from -10 to 55 °C.

3.9.2.2.3 Class 3. Class 3 equipment shall be subjected to operating temperatures ranging from 0 to 50 °C.

3.9.2.2.4 Class 4. Class 4 equipment shall be subjected to operating temperatures ranging from 10 to 40 °C.

3.9.2.3 Humidity. Equipment shall conform to the specified performance and accuracy requirements for conditions where the RH is 5 to 95±5 percent in the temperature range of 10 to 30 °C and as specified herein for the applicable class. At temperatures below 10 °C, the humidity is uncontrolled, but the equipment shall conform to the specified performance requirements (after the specified warm-up period) and shall withstand the effects of humidity up to 100 percent.

3.9.2.3.1 Class 1. Class 1 equipment shall be subjected to conditions where the RH is 5 to 85±5 percent in the temperature range of 30 to 40 °C and where the RH is 5 to 60±5 percent in the temperature range above 40 °C.

3.9.2.3.2 Class 2, 3, and 4. Class 2, 3, and 4 equipment shall be subjected to conditions where the RH is 5 to 75±5 percent in the temperature range of 30 to 40 °C and where the RH is 5 to 45±5 percent in the temperature range above 40 °C (as appropriate for the specified class).

3.9.3 Altitude. Class 1, 2, and 3 equipment shall conform to the specified performance and accuracy requirements while operating at an altitude of 4600 meters and after returning from an altitude of 4600 meters when tested in accordance with 4.5.5.2. Class 4 equipment shall conform to the specified performance and accuracy requirements while operating at an altitude of 3000 meters and after returning from an altitude of 4600 meters when tested in accordance with 4.5.5.2.

3.9.4 Vibration.

3.9.4.1 Random vibration. Class 1, 2, 3, and 4 equipment shall conform to the specified performance and accuracy requirements when subjected to the random vibration conditions specified in table III in accordance with 4.5.5.3.1.

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TABLE III. Random vibration by class.

Class	Duration per axis (minutes)	Frequency (Hz)	Slope (dB/Octave)	PSD (g^2/Hz)
1	60	10-1000 1000-2000 ≥2000	0 -6 ---	0.04 --- 0.01
2	30	10-500	0	0.03
3, 4	10	5-100 100-137 137-350 350-500 ≥500	0 -6 0 -6 ---	0.015 --- 0.0075 --- 0.0039

3.9.4.2 Sinusoidal vibration. When specified (see 6.2), class 1, 2, 3, and 4 equipment shall conform to the specified performance and accuracy requirements when subjected to the sinusoidal vibration conditions specified in table IV in accordance with 4.5.5.3.2.

TABLE IV. Sinusoidal vibration by class.

Class	Frequency increments (Hz)	Displacement peak-to-peak (mm)	Resonance search time per increment (minutes 1/)	Cycling time per axis (minutes)	Resonance dwell time per axis (minutes)
1, 2	5 to 15 16 to 25 26 to 55	1.5 1.0 0.5	5 5 5	15	10
3, 4	5 to 55	0.33	15	15	10

NOTE:

1/ Search times specified shall be for a low frequency to high frequency to low frequency sweep of the applicable increment.

3.9.4.3 Loose cargo bounce. Unless other classes of equipment are specified (see 6.2), only class 1 and 2 equipment shall conform to the specified performance and accuracy requirements when subjected to the loose cargo bounce conditions in accordance with 4.5.5.3.3.

3.9.5 Mechanical shock.

3.9.5.1 Functional shock. Class 1, 2, 3, and 4 equipment shall conform to the specified performance and accuracy requirements when subjected to the 30g half-sine shock test of 4.5.5.4.1.

3.9.5.2 Transit drop. Equipment for which the enclosure acts as the transit case, equipment that is specified to include a transit case, and all class 1 equipment shall conform to the specified performance and accuracy requirements when subjected to the transit drop test of 4.5.5.4.2. When a soft transit case is specified (see 6.2), only those with shoulder straps shall be subjected to this test. The equipment shall remain inside the transit case throughout this test.

3.9.5.3 Bench handling. Equipment shall conform to the specified performance and accuracy requirements and there shall be no damage to any controls, indicators, or fuseholders when subjected to the bench handling test of 4.5.5.4.3.

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3.9.5.4 High-impact shock. When specified (see 6.2), equipment shall conform to the specified performance and accuracy requirements when subjected to a high-impact shock test. The high-impact shock test shall be as specified (see 6.2).

3.9.6 Water resistance. Water resistance shall be as specified herein and summarized in table V.

3.9.6.1 Watertight.

3.9.6.1.1 Class 1. Class 1 equipment shall conform to the specified performance and accuracy requirements when subjected to the 0.9-meter submergence test in accordance with 4.5.5.5.1.

3.9.6.1.2 Class 2, 3, and 4. Unless otherwise specified (see 6.2), class 2, 3, and 4 equipment need not be watertight.

3.9.6.1.3 Airtight transit case. When a transit case is specified, and it is specified to be airtight (see 6.2), the transit case shall be watertight. Equipment housed in an airtight transit case shall conform to the specified performance and accuracy requirements when subjected to the 0.9-meter submergence test with the equipment in the transit case in accordance with 4.5.5.5.1.

3.9.6.2 Splash-proof.

3.9.6.2.1 Class 1. Class 1 equipment shall conform to the specified performance and accuracy requirements after covers, if present, have been removed when subjected to the splash-proof test in accordance with 4.5.5.5.2.

3.9.6.2.2 Class 2, 3, and 4. Unless otherwise specified (see 6.2), class 2, 3, and 4 equipment need not be splash-proof.

3.9.6.3 Drip-proof.

3.9.6.3.1 Class 2, 3, and 4. Unless otherwise specified (see 6.2), class 2, 3, and 4 equipment need not be drip-proof. This requirement is not applicable for class 1 equipment.

3.9.6.3.2 Hard transit case. When a hard transit case is specified (see 6.2), the transit case shall conform to the specified performance and accuracy requirements when subjected to the drip-proof test with the equipment in the transit case in accordance with 4.5.5.5.3. Testing of this requirement is not applicable for airtight transit cases.

3.9.6.3.3 Soft transit case. Unless otherwise specified (see 6.2), if a soft transit case is specified, it need not be drip-proof.

TABLE V. Water resistance.

Requirement	Class 1	Class 2, 3, and 4	Transit case – general	Transit case – airtight	Transit case – hard	Transit case – soft
Watertight	Submergence 0.9 m	1/	1/	Submergence 0.9 m	1/	N/A
Splash-proof	Yes	1/	N/A	N/A	N/A	N/A
Drip-proof	N/A	1/	1/	N/A	Yes	1/

NOTE:
1/ When specified (see 6.2).

3.9.7 Fungus resistance. Equipment shall not contain materials that provide nutrients for the growth of fungus. When a transit case is specified (see 6.2), it shall not contain materials that provide nutrients for the growth of fungus. Verification of fungus resistance and fungus-inert material shall be in accordance with 4.5.6.1.

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3.9.8 Salt atmosphere.

3.9.8.1 Enclosure exposure. Equipment shall conform to the specified performance and accuracy requirements when subjected to the salt fog test in accordance with 4.5.6.2.1.

3.9.8.2 Structural exposure. Material used for the fabrication of panels and structural parts shall conform to all applicable requirements after being exposed for 48 hours to a 5-percent salt atmosphere in accordance with 4.5.6.2.2. Unless otherwise specified (see 6.2), testing samples of panels and structural parts for class 1 equipment shall be required.

3.9.9 Explosive atmosphere for class 1 equipment. The operation of class 1 equipment shall not cause ignition of an ambient-explosive-gaseous mixture with air when tested in accordance with 4.5.6.3.

3.9.10 Dust exposure for class 1 equipment and hard transit cases. Class 1 equipment and equipment in hard transit cases shall conform to all requirements after exposure to dust when tested in accordance with 4.5.6.4.

3.9.11 DC field (Navy) magnetic environment. When specified (see 6.2), the equipment shall conform to the performance and accuracy specified when tested in accordance with the 20-oersted DC test of 4.5.6.6.

3.9.12 Acoustic noise. Unless otherwise specified (see 6.2), equipment shall not generate acoustic noise in excess of 70 A-weighted decibels (dBA) sound pressure level (SPL) when measured at the operator and bystander positions in accordance with ISO 7779.

3.9.13 Solar radiation. When specified (see 6.2), class 1 and 2 equipment shall conform to applicable requirements after exposure to solar radiation when tested in accordance with 4.5.6.8.

3.10 Electromagnetic compatibility (EMC). EMC control for all classes of equipment shall meet the requirements in table VI when tested in accordance with 4.5.6.5.

TABLE VI. EMC requirements.

Description of requirement	Test limit
Conducted emissions, power leads	IEC CISPR 11 Ed. 4.1, class A, 150 kHz to 30 MHz
Conducted immunity, power leads	IEC 61000-4-6 Ed. 3.0, 3 V AC-RMS, 150 kHz to 80 MHz, 80% amplitude modulation with 1-kHz sine wave
Conducted immunity, power leads, spikes	IEC 61000-4-4 Ed. 2.0, 1-kV peak for AC lines, 0.5-kV peak for I/O signal, data, and control ports, 100-kHz repetition frequency
Radiated emissions, electric field	IEC CISPR 11 Ed. 4.1, class A, 30 MHz to 1 GHz 1/
Radiated immunity, electric field	IEC 61000-4-3 Ed. 3.1, 3 V/m, 80 MHz to 1000 MHz, 80% amplitude modulation with 1-kHz sine wave
Electrostatic discharge immunity	IEC 61000-4-2 Ed. 2.0, 1-kV to 4-kV contact discharge, 2-kV to 8-kV air discharge

NOTE:

1/ IEC CISPR 11, class A, 30 MHz to 1000 MHz, 0.5-kV peak for signal, control, and DC lines, 5 nS Tr (T-rise) and 50 nS Th (T-hold), 5K.

3.11 Safety. The equipment shall meet the safety requirements of IEC 61010-1. For safety, all accessible surfaces of the equipment shall be at ground potential. The power cable shall include a safety ground conductor. Disconnecting the safety ground shall also disconnect the power conductors for drawer assemblies.

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3.12 Marking and identification.

3.12.1 Marking. Test equipment shall be marked in a manner that does not adversely affect safety or performance.

3.12.1.1 Reference designations. Reference designations shall identify each part for its particular circuit application. Subminiaturized and nonrepairable assemblies (see 6.6.17) need not be marked with reference designations. Reference designator usage shall be in accordance with IEEE 315.

3.12.1.2 Warning markings. Warning markings shall warn of the location, the nature, and the extent of a hazard. Letters of warning markings shall be of clearly legible gothic capitals. Warning markings shall have high contrast between the letter and background colors. The markings shall be as permanent as normal life expectancy of the test equipment and located as close as possible to the point of danger. Warning markings indicating circuits of more than 500 VDC or Vrms shall read as follows:

DANGER HIGH VOLTAGE (maximum applicable voltage) VOLTS

Markings shall be in accordance with IEC 61010-031, which requires the following for warning and caution markings:

a. The precautionary signal word shall be at least 2.5 mm high.

b. The text shall be at least 1.5 mm high and contrasting in color to the background.

c. If molded or stamped in a material, the text shall be at least 2.0 mm high and, if not contrasting in color, a depth or raised height of at least 0.5 mm.

3.12.1.3 Battery warning label. Equipment containing batteries shall be provided with a battery warning label attached to the equipment. Battery warning labels shall be as follows:

WARNING REMOVE BATTERIES BEFORE SHIPMENT OR INACTIVE STORAGE OF 30 DAYS OR MORE NOTE: DO NOT REMOVE INTERNAL MEMORY BACKUP BATTERIES

3.12.1.4 Panel markings and processes. Panel markings shall enable the operator to identify the functions and the operations of all items requiring operator intervention.

3.12.2 Equipment identification. Equipment identification shall include the following items and be as specified herein:

a. Manufacturer name

b. Equipment model number

c. Equipment name

d. Place of manufacture

e. Serial number

3.12.2.1 Identification plate. An identification plate shall be permanently affixed to the equipment and shall contain the following information:

a. Joint Electronics Type Designation Automated System (JETDAS) nomenclature (when assigned)

b. Manufacturer/Commercial and Government Entity (CAGE) code

c. Part number

d. Serial number

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e. National stock number (NSN)

f. Contract number

3.12.2.2 Identification plate supplemental information. When accessories are specified (see 6.2) and the combination has a part or model number designation that differs from that of the main unit, the identification plate shall also contain the following information:

a. Part number of main unit and all accessories (excluding cables and adapters) comprising the equipment configuration.

b. Model configuration number.

Whether a configuration model designation is assigned or not, all major accessories shall be marked as follows:

c. Part of number (p/o): Model number assigned to the equipment configuration or main unit model number, as applicable.

d. Serial number (s/n): Serial number assigned to the equipment comprising the set or the serial number of the primary instrument if a separate s/n is not assigned to the set.

3.12.3 Transit case identification plate. When a transit case is specified (see 6.2), the transit case shall include a permanently affixed identification plate that contains the information specified in 3.12.2. When specified (see 6.2), the identification plate shall also contain the information specified in 3.12.2.1. The location of the identification plate on the transit case shall be as specified (see 6.2).

3.12.4 Reusable pouch or container. When specified (see 6.2), a reusable pouch or container shall be provided. The pouch or container shall contain accessories such as sensors, leads, bridges, loads, and adapters. Containers shall be marked “accessories for use with” (test equipment nomenclature or model number) or “maintenance aids for use with” (test equipment nomenclature or model number), as applicable. If assigned, the test equipment nomenclature shall be used in lieu of the model number.

3.12.5 Nomenclature assignment. When specified (see 6.2), the nomenclature assignment will be provided by the Government.

3.13 Calibration interval. The calibration interval (see 6.6.7) shall be 12 months minimum. The equipment shall be within all accuracy requirements specified herein with an 85 percent or greater confidence factor following its full calibration interval.

3.14 Reliability. Class 1 equipment shall have a statistical certainty of 99 percent, class 2 equipment shall have a 95 percent statistical certainty, and class 3 and 4 equipment shall have a 90 percent statistical certainty. (This allows for infant mortality, random variation, and spurious failures.) Unless otherwise specified (see 6.2), the following minimum equipment reliability requirements shall be met:

a. Handheld equipment. The design of the equipment, except for COTS non-rechargeable batteries, shall be such that under normal use and operation, the equipment does not fail within 2,500 hours of operation.

b. Bench top equipment. The design of the equipment shall be such that, under normal use and operation, the equipment does not fail within 10,000 hours of operation (2,000 power on hours [POH] for 5 years or a 23 percent duty cycle).

c. Rack mount equipment or any other test equipment that has a 100 percent duty cycle. The design of the equipment shall be such that, under normal use and operation, the equipment does not fail within 40,000 hours of operation (8,000 POH for 5 years or a 91 percent duty cycle).

3.15 Maintainability.

3.15.1 Fault isolation. The design of the equipment shall permit isolation of faults and repair down to the component item or lowest replaceable unit with the maintenance provisions furnished.

3.15.2 Preventive maintenance. Preventive maintenance shall not require more than 15 minutes per a 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed.

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3.15.3 Maintenance provisions. Equipment shall be as follows to facilitate maintenance:

a. Accessibility for maintenance. Equipment shall be designed to ensure that maintenance access can be accomplished utilizing ordinary tools (see 6.6.33). The equipment shall be constructed so that no damage to any component shall occur and no permanent distortion to any structural member shall be caused during maintenance and calibration.

b. Maintenance and calibration aids. When specified (see 6.2), circuit board extenders, special adapters, special tools (see 6.6.26), and patch cables required for maintenance or calibration shall be provided. They shall be identified as a set with a unique part number and NSN. Stowage shall be in a reusable pouch or container.

c. Accessibility for repair. Equipment shall be designed so that subassemblies and chassis components can be removed without removing other hardwired subassemblies or components. Printed circuit cards shall be removable without the need to unsolder cables and interconnecting wiring. Printed circuit cards designed primarily to distribute power and signals to other printed cards are excluded from this requirement. When motherboards are used, they shall be accessible from both sides to facilitate troubleshooting or repair.

d. Accessibility for adjustment. Equipment design shall permit adjustments to be made without removing any component, printed circuit cards, or subassembly. The use of extender cards is permitted.

3.15.4 Washability. When specified (see 6.2), equipment shall conform to all requirements after being washed and there shall be no evidence of deterioration when tested in accordance with 4.5.7.1.2.

3.16 Environmental stress screening (applicable to class 1 only). When specified (see 6.2), equipment shall withstand environmental stress screening with no damage, loss of functionality, or degradation of performance. The applicable environmental stress screening requirements shall be as follows:

a. Random vibration. Random vibration shall be performed in accordance with 4.5.3.1.c.

b. Temperature cycling. Temperature cycling shall be performed in accordance with 4.5.3.1.d.

3.17 Use and installation manual and maintenance and service manual. The manufacturer's Use and Installation Manual (see 6.6.32) and the Maintenance and Service Manual (see 6.6.15) shall be as specified (see 6.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Bid sample inspection (see 4.2 and 6.4).

b. First article inspection (see 4.3 and 6.3).

c. Conformance inspection (see 4.4 and 6.5).

4.2 Bid sample inspection. When specified (see 6.2), the bid sample shall consist of two units, unless otherwise specified (see 6.2), with accessories, the Use and Installation Manual, and a clear and concise rationale showing how the reliability and maintainability characteristics of the equipment comply with the requirements of this specification. The bid sample shall be representative of equipment that complies with the requirements of this specification. Bid sample inspections are classified into groups in accordance with table VIII and appendix A.

4.3 First article inspection. First article inspection shall be performed on up to five units, unless otherwise specified (see 6.2), when a first article sample is required (see 3.1.2 and 6.2). First article inspections are classified into groups in accordance with table VIII and 4.3.1 through 4.3.6. First article inspection shall be performed on the units as specified in 4.5 and as follows:

a. Failure criteria. Failure (see 6.6.10) of any group A through group E test shall be considered a major defect. Failure of group F tests shall result in elimination from further consideration for procurement.

b. Test sequence. The test sequence shall be as specified herein for groups A through F. If group B testing is not specified (see 6.2), the test sequence shall proceed from group A to groups C and D to group E. Group D testing may be performed concurrently with, or sequentially after, group C testing.

c. Test conditions and control settings. The test conditions and control settings shall be in accordance with 4.5.1.

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4.3.1 Group A. The five units shall be subjected to the group A inspection specified in 4.5.3.

4.3.2 Group B. The five units that have passed group A inspection (see 4.3.1) shall be subjected to the group B tests of 4.5.4. If group B testing is not specified (see 6.2), the test sequence shall proceed from group A to groups C and D, as applicable.

4.3.3 Group C. Three units that have passed group B testing (see 4.3.2) shall be subjected to the group C tests of 4.5.5.

4.3.4 Group D. Two units that have passed group C testing (see 4.3.3) shall be subjected to the group D tests of 4.5.6.

4.3.5 Group E. Units that have passed group D testing (see 4.3.4) shall be subjected to the group E tests of 4.5.7. When the number of units required for group E testing exceeds the number required for group D testing, the additional units shall be selected from the units that have passed group C testing. The number of units tested shall be as follows:

a. Maintainability. One unit shall be tested. When the washability test is specified (see 6.2), the test shall be performed on the unit prior to the maintainability demonstration.

b. Acoustic noise. One unit shall be tested. Equipment that does not contain a motor, such as a fan or blower, or other audible noise source shall not be tested.

c. Dimensions and weight. Two units shall be tested.

d. Mechanical stability. Two units shall be tested. Units smaller than 2000 cc with a weight less than 2.5 kg shall not be tested.

e. Equipment emanations. Two units shall be tested. X-radiation tests shall not be performed unless the equipment is designed to have operating potentials of 10 kilovolts (kV) or greater or could have potentials of 10 kV or greater due to the failure of any one component.

f. Additional tests. Additional group E tests shall be as specified (see 6.2).

4.3.6 Group F. Group F tests shall be performed by evaluating the manufacturer's data to establish that reliability falls within the parameters outlined in 3.14.

4.4 Conformance inspection. Conformance inspections are classified into groups in accordance with table VIII and 4.4.1 through 4.4.6. Conformance inspection shall consist of groups A and B. The Government reserves the right to invoke groups C, D, E, and F for conformance inspection.

a. Failure criteria. Failure of any group test shall be considered a major defect. All units in the sample size shall be rejected. For the first failure, corrective action shall be implemented, and the lot samples shall be resubmitted for conformance inspection. If the resubmitted lot fails, it is at the Government's discretion to reject the lot samples or to cancel the current contract.

b. Test sequence. The test sequence shall be as specified herein for groups A through E prior to the start of the group F verification test of 4.4.6. If group B testing is not specified (see 6.2), the test sequence shall proceed from group A to groups C and D to group E. Group D testing may be performed concurrently with, or sequentially after, group C testing.

c. Test conditions and control settings. The test conditions and control settings shall be in accordance with 4.5.1.

4.4.1 Group A. Each unit shall be subjected to the group A inspection specified in 4.5.3 to ensure proper functioning of the equipment, including all operating controls, and conformance to the safety requirements.

4.4.2 Group B. The units that have passed group A inspection (see 4.4.1) shall be subjected to the group B tests of 4.5.4. If group B testing is not specified (see 6.2), the test sequence shall proceed from group A to groups C and D, as applicable.

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4.4.2.1 Sampling. A random sample of units from each lot (see 6.6.14) shall be selected for group B testing in accordance with table VII. If a defect is found in any sample, the entire lot shall be rejected. When group D testing is to be performed concurrently with group C testing, or when required for additional group E tests, the group B sample size shall be increased to provide the required samples for group C, D, and E testing.

4.4.3 Group C. Units that have passed group B testing (see 4.4.2) shall be subjected to the group C tests of 4.5.5.

4.4.3.1 Sampling. A random sample of units from each lot shall be selected for group C testing in accordance with table VII. If a defect is found in any sample, the entire lot shall be rejected.

4.4.4 Group D. Units that have passed group C testing (see 4.4.3) shall be subjected to the group D tests of 4.5.6.

4.4.4.1 Sampling. A random sample of units from each lot shall be selected for group D testing in accordance with table VII. If a defect is found in any sample, the entire lot shall be rejected.

4.4.5 Group E. A random sample of units from the first production lot that has passed group D testing (see 4.4.4) shall be selected and subjected to the group E tests of 4.5.7. The number of units tested shall be as follows:

a. Maintainability. One unit shall be randomly selected and tested.

b. Acoustic noise. One unit shall be randomly selected and tested. Equipment that does not contain a motor, such as a fan or blower, or other audible noise source shall not be tested.

c. Dimensions and weight. Four units shall be randomly selected and tested.

d. Mechanical stability. One unit shall be randomly selected and tested. Units smaller than 2000 cc with a weight less than 2.5 kg shall not be tested.

e. Equipment emanations. One unit shall be randomly selected and tested. X-radiation tests shall not be performed unless the equipment is designed to have operating potentials of 10 kV or greater or could have potentials of 10 kV or greater due to circuit malfunction resulting from the failure of any component, including related sequential failures.

f. Additional tests. Additional group E tests shall be as specified (see 6.2).

4.4.6 Group F. Group F tests shall be performed by evaluating the manufacturer’s data to establish that reliability falls within the parameters outlined in 3.14.

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TABLE VII. Sampling for conformance inspection.

Lot Size	Group B	Group C	Group D
2 to 8	All	3	2
9 to 15	8	3	3
16 to 25	8	3	3
26 to 50	8	5	5
51 to 90	8	6	5
91 to 150	12	7	6
151 to 280	19	10	7
281 to 500	21	11	9
501 to 1,200	27	15	11
1,201 to 3,200	35	18	13
3,201 to 10,000	38	22	15
10,001 to 35,000	46	29	15
35,001 to 150,000	56	29	15

TABLE VIII. Bid sample, first article, and conformance inspection examinations and tests.

Examination or test	Requirement paragraph	Inspection paragraph
Group A:		
Pre-operational inspection	3.2 through 3.2.5, 3.4 through 3.8	4.5.3.1
Level A performance (see 6.6.19.1)	3.3	4.5.3.2
Leakage current	3.11	4.5.3.2.1
Group B:		
Level B performance (see 6.6.19.1)	1/	4.5.4 2/
Group C:		
T/H	3.9.2	4.5.5.1
T/H not operating	3.9.2.1	4.5.5.1.1
Temperature operating	3.9.2.2	4.5.5.1.1
Altitude	3.9.3	4.5.5.2
Vibration	3.9.4	4.5.5.3
Random vibration	3.9.4.1	4.5.5.3.1
Sinusoidal vibration	3.9.4.2	4.5.5.3.2 2/
Loose cargo bounce	3.9.4.3	4.5.5.3.3
Mechanical shock	3.9.5	4.5.5.4
Functional shock	3.9.5.1	4.5.5.4.1
Transit drop	3.9.5.2	4.5.5.4.2 3/
Bench handling	3.9.5.3	4.5.5.4.3
High-impact shock	3.9.5.4	4.5.5.4.4 1/, 2/

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TABLE VIII. Bid sample, first article, and conformance inspection examinations and tests – Continued.

Examination or test	Examination or test	Examination or test
Water resistance	3.9.6	4.5.5.5
Watertight	3.9.6.1	4.5.5.5.1 4/
Splash-proof	3.9.6.2	4.5.5.5.2 5/
Drip-proof	3.9.6.3	4.5.5.5.3 2/
Electrical power	3.5	4.5.5.6
Input power consumption	3.6.1.1	4.5.5.6.1
Voltage and frequency variation	3.6.1.2.1	4.5.5.6.2
Voltage-transient	3.6.1.2.2	4.5.5.6.3
Frequency-transient	3.6.1.2.2	4.5.5.6.4
Power source interruption	3.6.1.2.3	4.5.5.6.5
Verification, group C	3.9.1	4.5.5.7
Group D:		
Fungus resistance	3.9.7	4.5.6.1 2/
Salt fog	3.9.8	4.5.6.2 2/
Enclosure exposure (salt fog)	3.9.8.1	4.5.6.2.1 2/
Structural parts exposure (salt fog)	3.9.8.2	4.5.6.2.2 6/
Explosive atmosphere	3.9.9	4.5.6.3 7/
Dust	3.9.10	4.5.6.4 7/
EMC	3.10	4.5.6.5
Magnetic environment	3.9.11	4.5.6.6 2/
Verification, group D	3.9.1	4.5.6.7
Solar radiation	3.9.13	4.5.6.8 2/, 8/
Group E:		
Maintainability	3.15.1 through 3.15.4	4.5.7.1
Washability	3.15.4	4.5.7.1.2 2/
Acoustic noise	3.9.12	4.5.7.2
Dimensions	3.7.1	4.5.7.3
Weight	3.7.2	4.5.7.3
Mechanical stability	3.7.3	4.5.7.4
Equipment emanations	3.11	4.5.7.5 2/
Group F:		
Reliability	3.14	4.5.8 2/

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TABLE VIII. Bid sample, first article, and conformance inspection examinations and tests – Continued.

NOTES:
1/ To be performed as specified (see 6.2).
2/ To be performed only when specified (see 6.2).
3/ Required for class 1 (see 4.5.5.4.2).
4/ Required for class 1 and airtight transit cases.
5/ Required for class 1 with covers removed.
6/ Required for class 1 (see 4.5.6.2.2).
7/ Class 1 only and other classes only when specified (see 6.2).
8/ Class 1 and 2 only.

4.5 Examination and test methods. The test methods, procedures, conditions, and failure criteria specified herein shall be applied. Equipment operation shall be verified as follows:

a. Satisfactory Operation Check (SOC). The SOC (see 6.6.23) is a minimal test to verify that the equipment is operating normally. When an SOC is required, the parameters and the limits of acceptability shall be as specified (see 6.2).

b. Satisfactory Operation Test (SOT). The SOT (see 6.6.24) is a test to verify that the equipment is operating normally and that it conforms to salient operational parameters for a limited number of verification tests. When an SOT is required, the parameters and the limits of acceptability shall be as specified (see 6.2).

4.5.1 Conditions and control settings.

4.5.1.1 Test conditions. Unless otherwise specified herein, the examination and test methods of 4.5 shall be performed under the following conditions:

a. Temperature: 25±10 °C.

b. Humidity: 20 to 70 percent RH.

c. Altitude: 0 to 4600 meters, except as noted in table II, note 8.

d. Power: Nominal or alternate power source specified for the equipment.

e. Attitude: Normal operating position.

Ambient conditions within the specified ranges shall not be controlled. For equipment having a specified warm-up or stabilization period after turn-on, formal measurements and observations shall be made only after the specified interval (see 3.9.1).

4.5.1.1.1 Tolerance of test conditions. The tolerance of test conditions shall be as follows:

a. Temperature. The test item shall be totally surrounded by an envelope of air except at necessary support joints. The temperature of the test section measurement system and the temperature gradient throughout this envelope, which is measured close to the test item, shall be within ±2 °C of the test temperature and shall not exceed 1 °C per meter or a maximum of 2.2 °C total (equipment not operating).

b. Pressure. Pressure shall be measured with an accuracy of ±5 percent of the measured value.

c. Humidity. RH at the chamber control sensor shall be ±5 percent RH of the measured value.

d. Acceleration. Acceleration shall be measured to within ±10 percent.

e. Time. Elapsed time shall be measured with an accuracy of ±1 percent.

4.5.1.1.2 Accuracy of test instrument calibration. All measurements of the test conditions shall be made with accurate instruments. The accuracy of test instruments and test equipment used to control or monitor the test parameters shall be verified prior to and following each test and then calibrated at predetermined intervals.

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4.5.1.2 Control settings. The control settings shall be as determined by the approved equipment test procedure (see 6.6.2).

4.5.2 Performance of test.

4.5.2.1 Pre-test qualification. Prior to proceeding with the environmental tests, the test item shall be operated under standard ambient conditions (see 4.5.1.1) to evaluate the performance characteristics of the equipment. This test is used to establish the level of performance of the equipment at the outset of testing prior to any environmental tests. This test shall be performed before, during, and after the environmental tests, whenever an SOT is required. Degradation of the equipment performance shall be noted if it exceeds any bound herein or as specified (see 6.2).

4.5.2.2 Installation of test item in test facility. The test item shall be installed in the test facility in a manner that simulates service usage, making connections and attaching instrumentation as necessary. Plugs, covers, and inspection plates not used in operation but used in servicing shall remain in place. When mechanical or electrical connections are not used, the connections normally protected in service shall be covered. For tests where temperature values are controlled, the test chamber shall be at standard ambient conditions when the test item is installed. The test item shall be operated to determine that no malfunction or damage was caused due to faulty installation or handling. The requirement for operation following installation of the test item in the test facility is applicable only when operation is required during exposure to the specified test.

4.5.2.3 Performance check during test. When operation of the test item is required during the test exposure, tests shall be performed to determine whether the test exposure is producing changes in performance when compared to the pre-test qualification baseline.

4.5.2.4 Post-test inspection. At the completion of each environmental test, the test item shall be inspected in accordance with the pre-test qualification (see 4.5.2.1).

4.5.2.5 Failure criteria. Failure criteria shall be as follows:

a. The item shall have failed the test if any of the following conditions occur:

(1) Monitored functional parameters deviate beyond the acceptable limits established in 4.5.2.1.

(2) Catastrophic or structural failure.

(3) Mechanical binding or loose parts, including screws, clamps, bolts, and nuts, that result in component failure or a hazard to personnel safety.

(4) Malfunction.

(5) Degradation of performance beyond the limits herein or as specified (see 6.2).

(6) Any additional deviations from acceptable criteria established before the test.

(7) Deterioration, corrosion, or change in tolerance limits of any internal or external parts that could prevent the test item from conforming to operational service or maintenance requirements.

b. Additional or different failure criteria shall be as specified (see 6.2).

4.5.2.6 Test equipment. Test equipment used for inspections and tests shall be as determined in the approved equipment test procedure. The test equipment shall be properly maintained and calibrated.

4.5.3 Group A inspection and tests. Group A inspection shall consist of the pre-operational inspection of 4.5.3.1 and the level A performance tests of 4.5.3.2.

4.5.3.1 Pre-operational inspection. Prior to the beginning of the level A test (see 4.5.3.2), the following pre-operational inspections shall be performed on all equipment:

a. Mechanical and visual examination. The equipment shall be given a thorough mechanical and visual examination to determine that all materials, workmanship, and safety characteristics comply with the specified requirements.

b. Electrical circuit configuration. The equipment shall be examined or tested to confirm that the wiring is correct. Where applicable, the tests shall include the following requirements:

(1) All intra-module wiring shall be tested to ensure correctness.

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(2) The module grounding system shall be examined or tested to ensure proper separation of shield, signal, and framework grounds as well as metal-to-metal contact for panels and components that serve as electromagnetic shields.

When specified for class 1 only (see 6.2), the following examinations shall be performed:

c. Environmental stress screening (random vibration) (class 1 only). Prior to conducting the temperature cycling, random vibration shall be performed on each unit. Random vibration may be performed at the module, drawer, end item level, or line replaceable unit (LRU). All hardware, including cables and connectors, shall be exposed to vibration. The vibration shall be random for an accumulated time of 1 hour in the axis most susceptible to vibration excitation. Alternatively, subject to procuring activity approval, the vibration may be pseudo-random or complex waveform vibration. All items being screened shall be hard-mounted (without shock isolators) and subjected to the vibration conditions on figure 1. The control accelerometer shall be located next to one of the mounting points of the item being screened. Equipment having a bandwidth not greater than 10 Hz for vibration frequencies up to 500 Hz and 100 Hz for vibration frequencies above 500 Hz shall be used for the control and analysis of the power spectral density (PSD). The item being screened shall be energized during vibration, and appropriate input signals shall be applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during the process shall be corrected and the screening resumed.

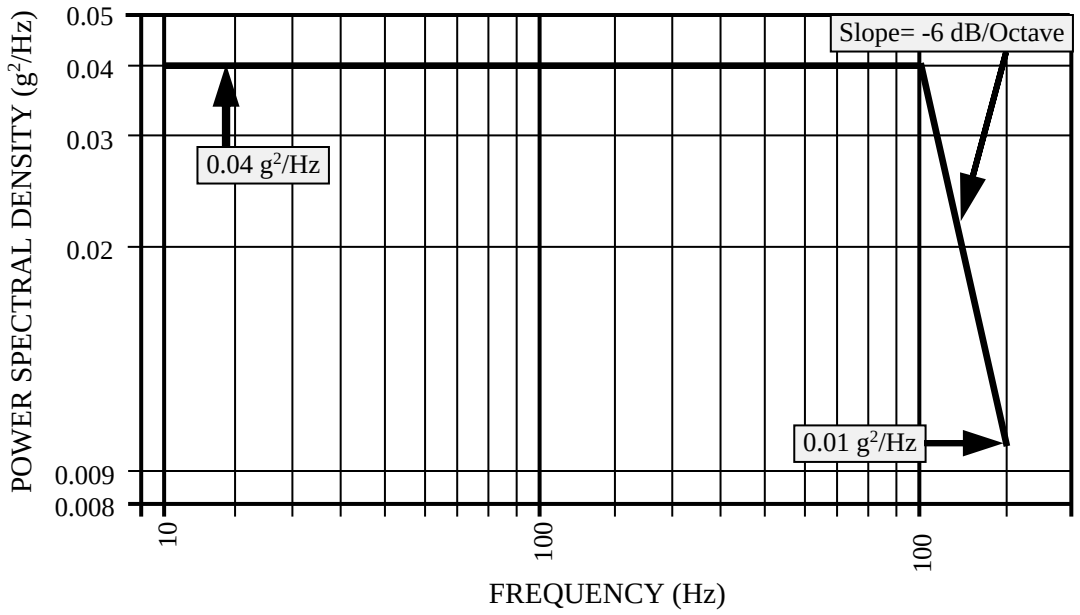


FIGURE 1. Random vibration environmental screening curve (for class 1 only).

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d. Environmental stress screening (temperature cycling) (class 1 only). Each unit shall be subjected to the temperature cycling shown on figure 2. The number of cycles required shall be determined by the electrical and electronic parts count in table IX. The temperature rate of change shall be not less than 5 °C per minute. Equipment power shall be turned on and off at the indicated times. When appropriate, the equipment drawers, panels, and enclosures shall be opened or removed for maximum exposure to the changing temperature. The last temperature cycle shall be failure-free. The dwell time shall be 80 percent of the time required for the largest electrical or electronic part to become temperature stabilized. Temperature stabilization shall be attained when the temperature of the part of the item being screened that is considered to have the longest thermal lag reaches a temperature within 2.0 °C of the specified temperature, except that any critical component shall be within 1 °C. Exceptions may occur in large items. When changing temperatures, the temperature of the chamber air may be adjusted up to 5 °C beyond the desired end point for a period of up to 1 hour to reduce stabilization time, provided that the stabilization requirements of this paragraph are ultimately attained relative to the specified end point temperature and provided the extended chamber temperatures will not cause damage to the item being screened. Failures occurring prior to the last original temperature cycle shall be repaired and the cycling continued. If failure occurs during the last original temperature cycle, table X shall be used to determine the number of additional cycles required.

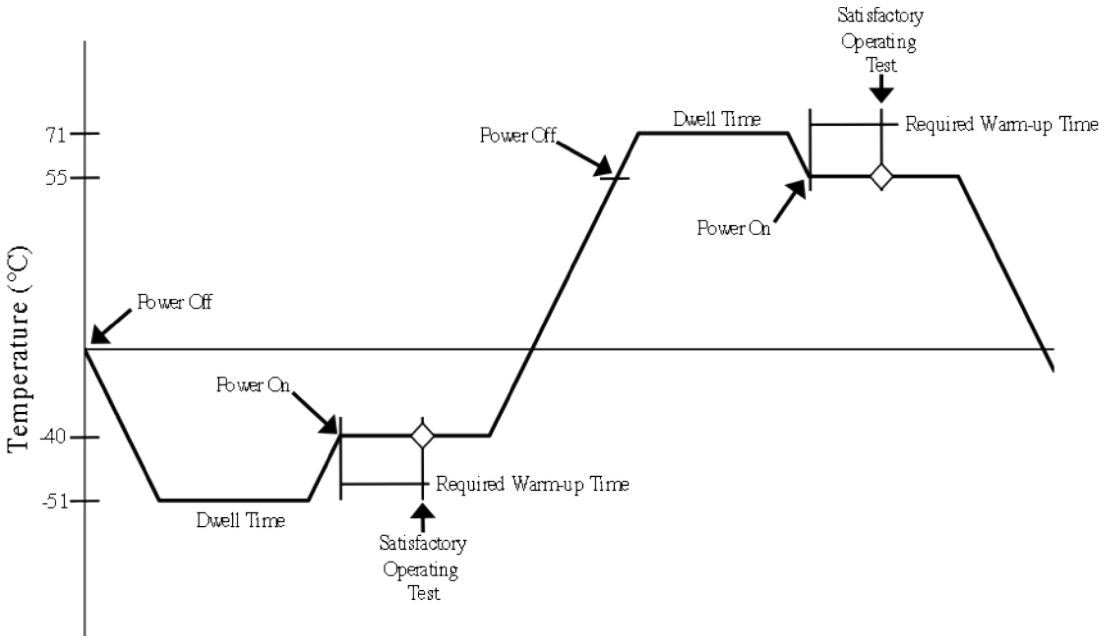


FIGURE 2. One cycle of the temperature screening curve (for class 1 only).

TABLE IX. Test cycles.

Electrical and electronic parts count	Number of temperature cycles
500 or less	3
501 to 1,000	4
1,001 to 2,000	6
2,001 to 3,000	8
3,001 or more	10

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TABLE X. Additional test cycles.

Percentage of total parts reworked or replaced 1/	Equipment parts count		
	500 or less	501 to 2,000	2,001 or more
0 to 0.1 percent	None	Repeat 1 cycle	Repeat 1 cycle
0.1 to 1 percent	Repeat 1 cycle	Repeat 1 cycle	Repeat 2 cycles
1 to 5 percent	Repeat 1 cycle	Repeat 2 cycles	Repeat 4 cycles
Over 5 percent	Repeat 2 cycles	Repeat 4 cycles	Repeat 6 cycles
	Last cycle shall be failure-free		

NOTE:
1/ Each reworked solder joint or resoldered lead shall count as one part replaced.

4.5.3.2 Level A performance tests. The level A performance tests for evaluating equipment performance shall be as required both herein and in the approved equipment test procedure, except that, when an approved equipment test procedure is not provided or authorized for use, conformance to performance requirements shall be verified. These tests shall be performed on each item of equipment and shall be conducted under the test conditions of 4.5.1.1.

4.5.3.2.1 Leakage current test. The leakage current test shall be in accordance with UL 60950-1 for class I, stationary, pluggable, type A equipment.

4.5.4 Group B tests. Group B tests consist of the level B performance tests and shall be performed as specified herein and in the approved equipment test procedure. These tests shall be performed and conducted under the test conditions of 4.5.1.1.

4.5.5 Group C tests. Group C tests shall be performed as specified herein and in the approved equipment test procedure. These tests shall be performed under the test conditions of 4.5.1.1.

4.5.5.1 Temperature and humidity (T/H) tests. The T/H tests shall be performed as specified herein. Figures 3 and 4 show the class 1 T/H tests. Figures 5 and 6 show the class 2 T/H tests. Figures 7 and 8 show the class 3 T/H tests. Figures 9 and 10 show the class 4 T/H tests. Transitions from one state to another shall not result in the creation of condensation.

4.5.5.1.1 T/H procedure. The test item shall be installed in the test facility in accordance with 4.5.2.2. During the tests specified in 4.5.5.1.1.1, the RH shall not be controlled. An RH of 95 percent (with the applicable tolerance) does not include conditions of precipitation. The rate of temperature change shall be 1 °C to 5 °C per minute. The temperature limits and RH are specified by class in table XI. Precipitation is not authorized during the T/H test.

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TABLE XI. Temperature limits.

Class	Temperature limits	
	Nonoperating equipment (°C)	**Operating equipment (°C at percent RH)**
1 1/	-51 to 71	-40 to 71 at 5 to 60 -40 to 40 at 5 to 85 -40 to 30 at 5 to 95
2	-51 to 71	-10 to 55 at 5 to 45 -10 to 40 at 5 to 75 -10 to 30 at 5 to 95
3	-40 to 71	0 to 50 at 5 to 45 0 to 40 at 5 to 75 0 to 30 at 5 to 95
4	-40 to 71	10 to 40 at 5 to 75 10 to 30 at 5 to 95

NOTE:
1/ For class 1 equipment, 20-minute intermittent operation (see 6.6.13) at 71 °C after the required warm-up period; 15 minutes not operating.

4.5.5.1.1.1 Temperature test procedure. The temperature test procedure consists of five independent tests (tests A through E) that can be performed in any sequence, except as indicated for test A. The profiles provided on figure 3 (for class 1), figure 5 (for class 2), figure 7 (for class 3), and figure 9 (for class 4) demonstrate only one possible sequence of testing. The detailed test procedure at the time of testing shall define the actual test sequence. The humidity during the test is uncontrolled for all tests except test D, where the humidity shall be controlled within the range of 5 to 20 percent RH with the applicable tolerance to simulate an arid environment. The testing may be interrupted after any test. Performance of the SOT shall occur at the end of each temperature test period, adding whatever time is required to perform the SOT. This means that the total, cumulative time required to perform temperature testing is the time required for each temperature test, the time required to perform an SOT at each temperature test, and any interruption period.

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TABLE XII. Temperature tests.

Test name	Test procedure
A	This test is the initial operation verification test and shall always be performed first in the test sequence. The test item shall be placed in the test chamber in accordance with 4.5.2.2. At room ambient temperature, the equipment shall operate for 2 hours, after which the SOT shall be performed.
B	The temperature shall be maintained at -40 °C (for class 1), -10 °C (for class 2), 0 °C (for class 3), or 10 °C (for class 4). The equipment shall not operate for 4 hours. Then, the test item shall be operated for the recommended warm-up period. The SOT shall be performed, and the results shall be compared with the test A results in accordance with 4.5.2.1. No alignment or adjustment of other than the operating controls shall be permitted throughout the test specified.
C	The temperature shall be maintained at -51 °C (for class 1 and 2) or -40 °C (for class 3 and 4). The equipment shall not operate for 14 hours. Following the 14-hour cold storage soak, the temperature shall be raised to 23 °C. For an additional 4 hours, the equipment shall be maintained at these conditions. Then, the test item shall be operated for the recommended warm-up period. The SOT shall be performed, and the results shall be compared with the test A results in accordance with 4.5.2.1. No alignment or adjustment of other than the operating controls shall be permitted throughout the test specified.
D	The humidity during this test shall be controlled within the range of 5 to 20% with the applicable tolerance. The temperature shall be maintained at 55 °C (for class 1 and 2), 50 °C (for class 3), or 40 °C (for class 4). The equipment shall operate for 4 hours. Following the 4-hour arid heat operating soak, the SOT shall be performed, and the results shall be compared with the test A results in accordance with 4.5.2.1. No alignment or adjustment of other than the operating controls shall be permitted throughout the test specified.
E (class 1)	The temperature shall be maintained at 71 °C. The equipment shall not operate for 14 hours. Then, the equipment shall be turned on and operated for 20 minutes at 71 °C after the required warm-up period. During the 20-minute operating period, the SOT (shortened as necessary to fit in the 20-minute operation period) shall be performed, and the results shall be compared with the test A results in accordance with 4.5.2.1. No alignment or adjustment of other than the operating controls shall be permitted throughout the test specified.
E (class 2, 3, and 4)	The temperature shall be maintained at 71 °C. The equipment shall not operate for 14 hours. Following the 14-hour hot storage soak, the temperature shall be lowered to 23 °C and the equipment shall be maintained at these conditions for an additional 4 hours. Then, the test item shall be operated for the recommended warm-up period. The SOT shall be performed, and the results shall be compared with the test A results in accordance with 4.5.2.1. No alignment or adjustment of other than the operating controls shall be permitted throughout the test specified.

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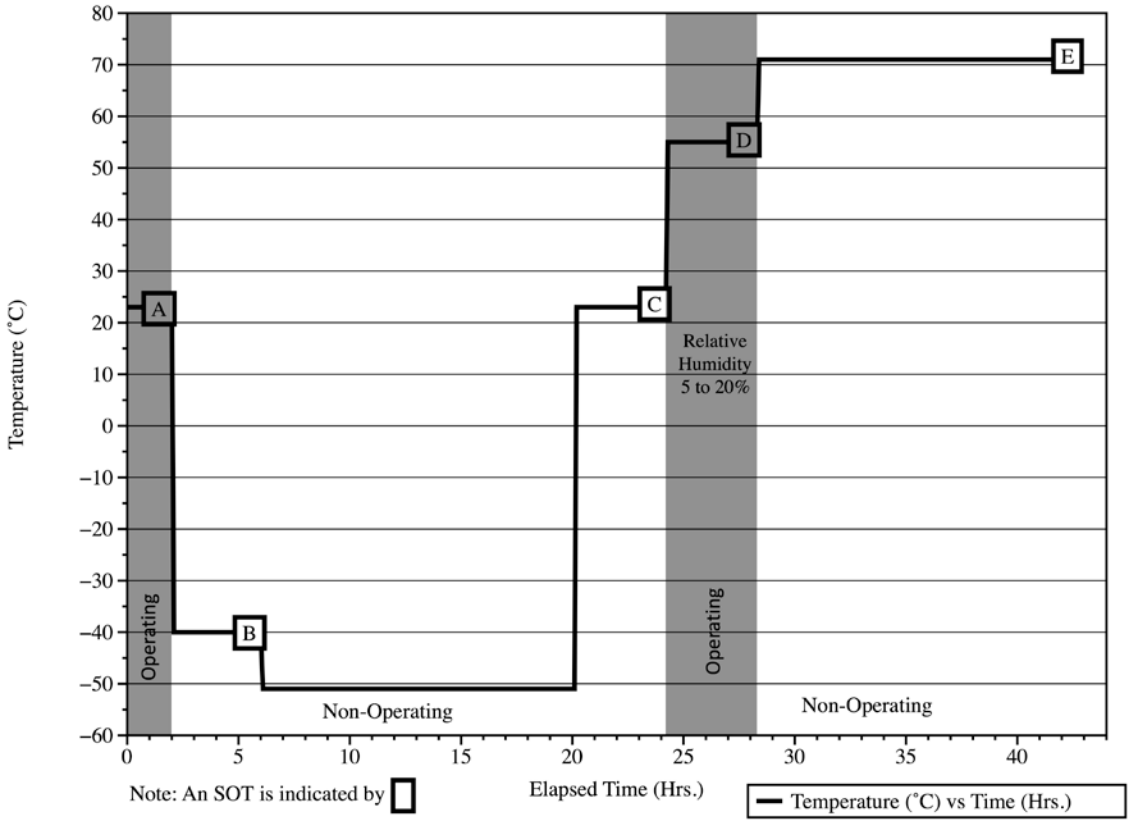


FIGURE 3. Class 1 sample temperature profile (including the arid climate test).

4.5.5.1.1.2 Humidity cycle procedure. The humidity cycle testing shall immediately follow after the testing of 4.5.5.1.1. This procedure shall consist of 5 days of temperature humidity cycling and each day's cycle shall consist of the profile displayed on figure 4 (for class 1), figure 6 (for class 2), figure 8 (for class 3), or figure 10 (for class 4). SOTs shall be performed at the times indicated on the figures as annotated by the rectangular symbols a through c. The following information applies:

a. An SOT at normal room ambient conditions shall be conducted prior to, and at the conclusion of, the 5-day humidity test.

b. During the humidity cycle, the equipment shall only operate during the warm-up period and the SOT.

c. The SOTs, as annotated by the rectangular symbols a, b, and c, shall be performed at least once each during the 5 days of humidity cycling at the indicated times. SOT a and b, as appropriate, shall be performed at any of the cycles 2, 3, 4, or 5. SOT c shall be performed at least at cycle 5. SOTs may also be performed at any or all cycles at the indicated times.

d. To accommodate varying times for completing SOTs, the cycle timing after a test may be adjusted to allow a return back to the regular profile timing. However, a minimum 4-hour dwell time prior to the period of operation shall be observed.

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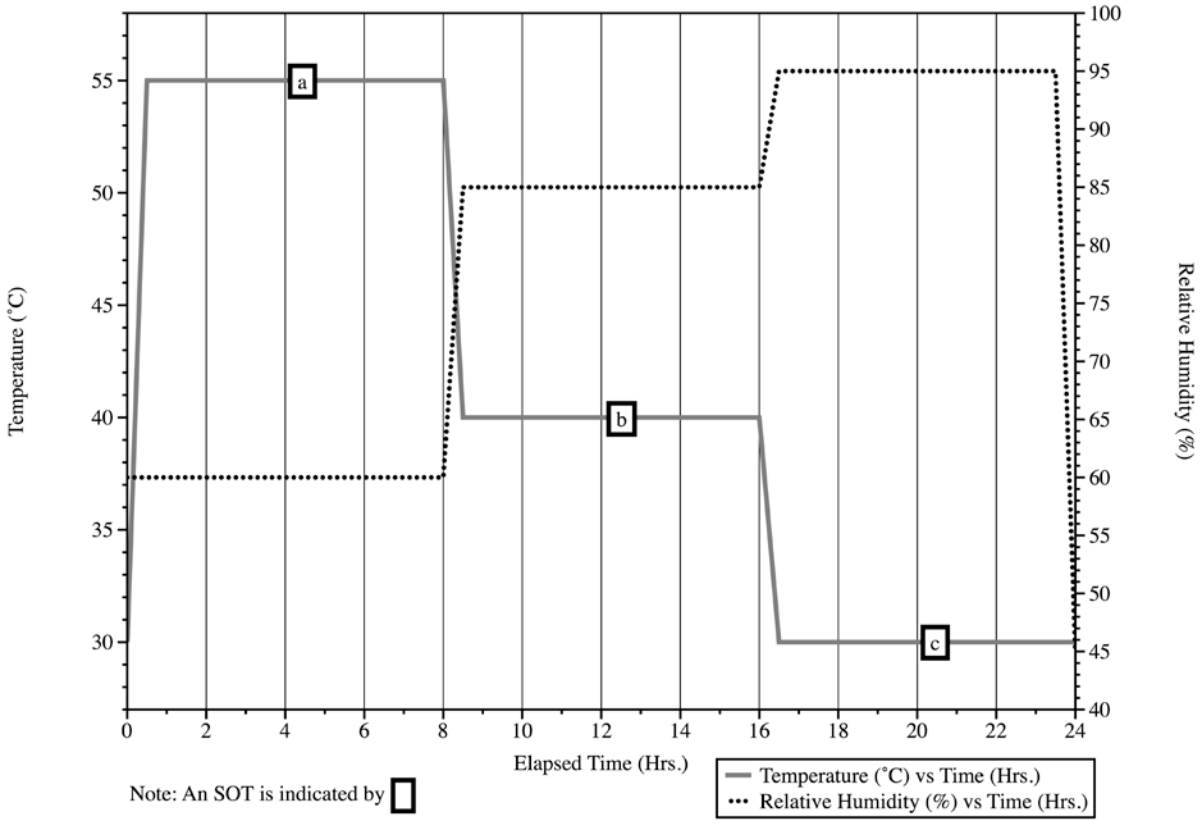


FIGURE 4. Class 1 sample humidity cycle profile.

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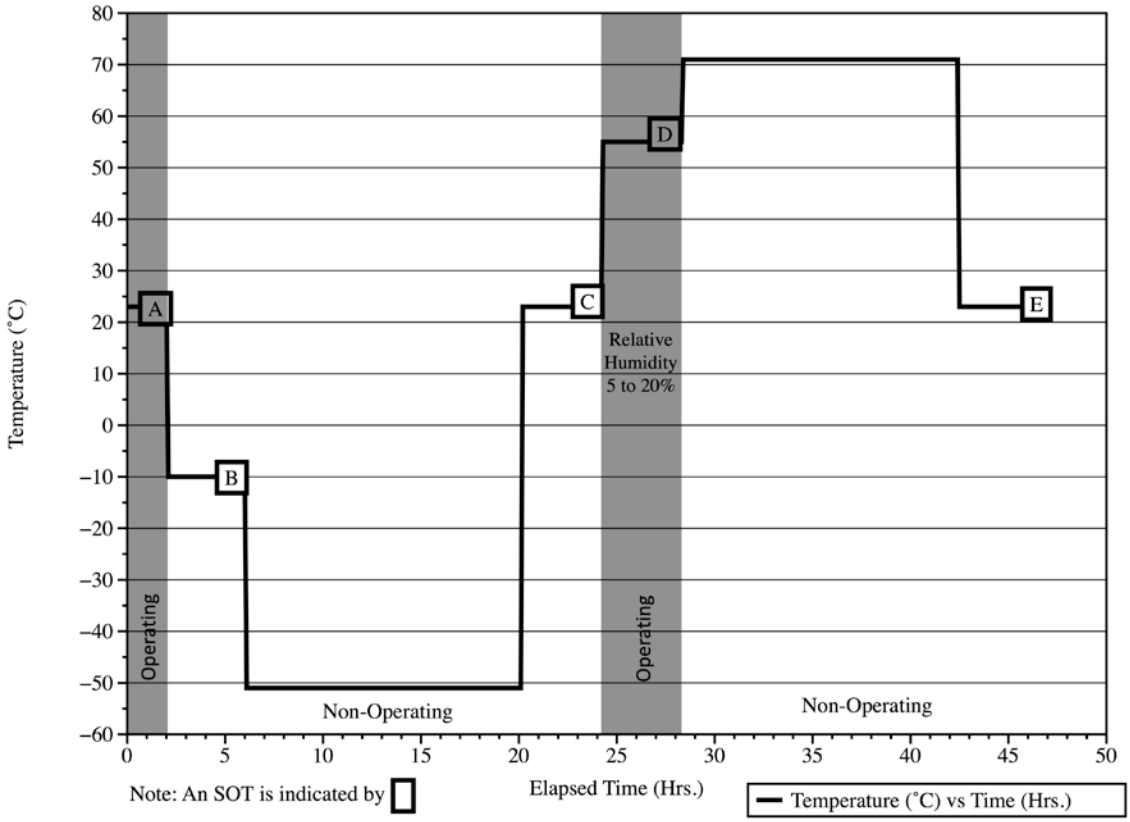


FIGURE 5. Class 2 sample temperature profile (including the arid climate test).

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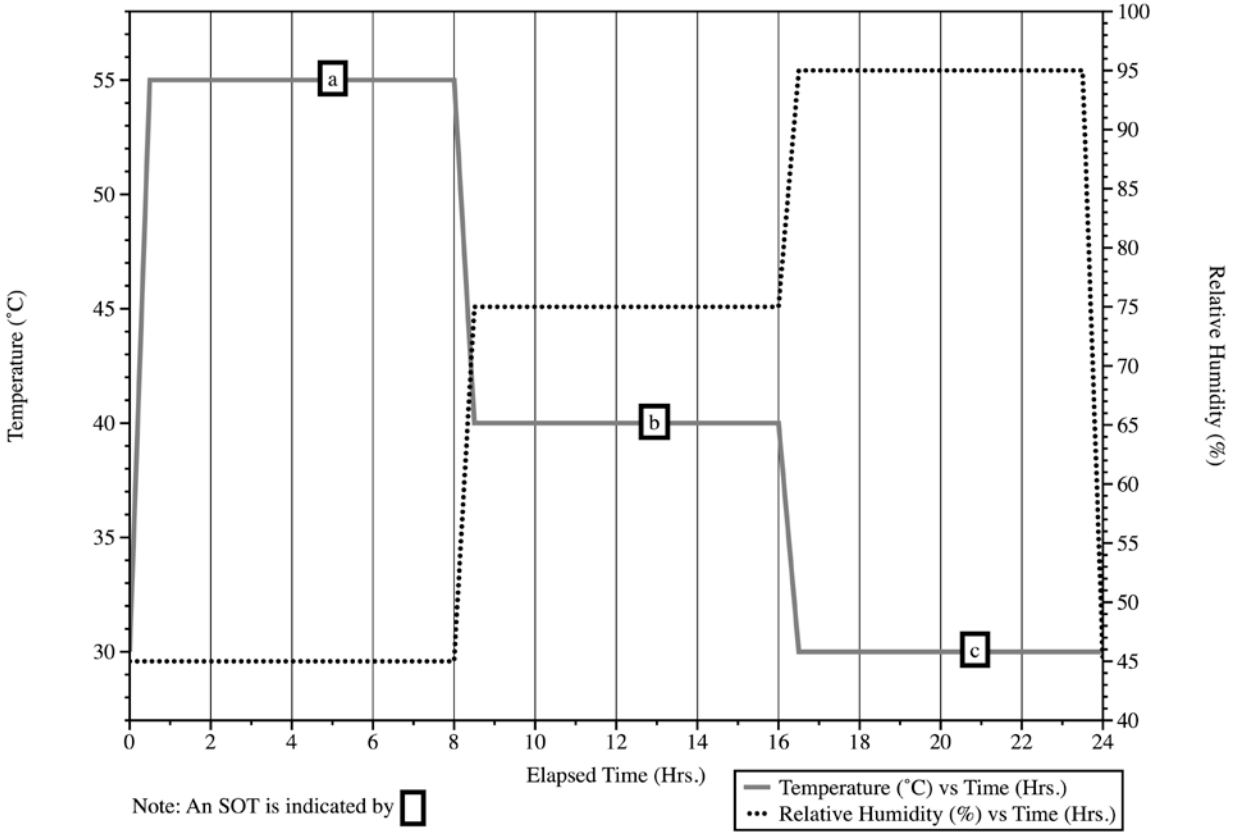


FIGURE 6. Class 2 sample humidity cycle profile.

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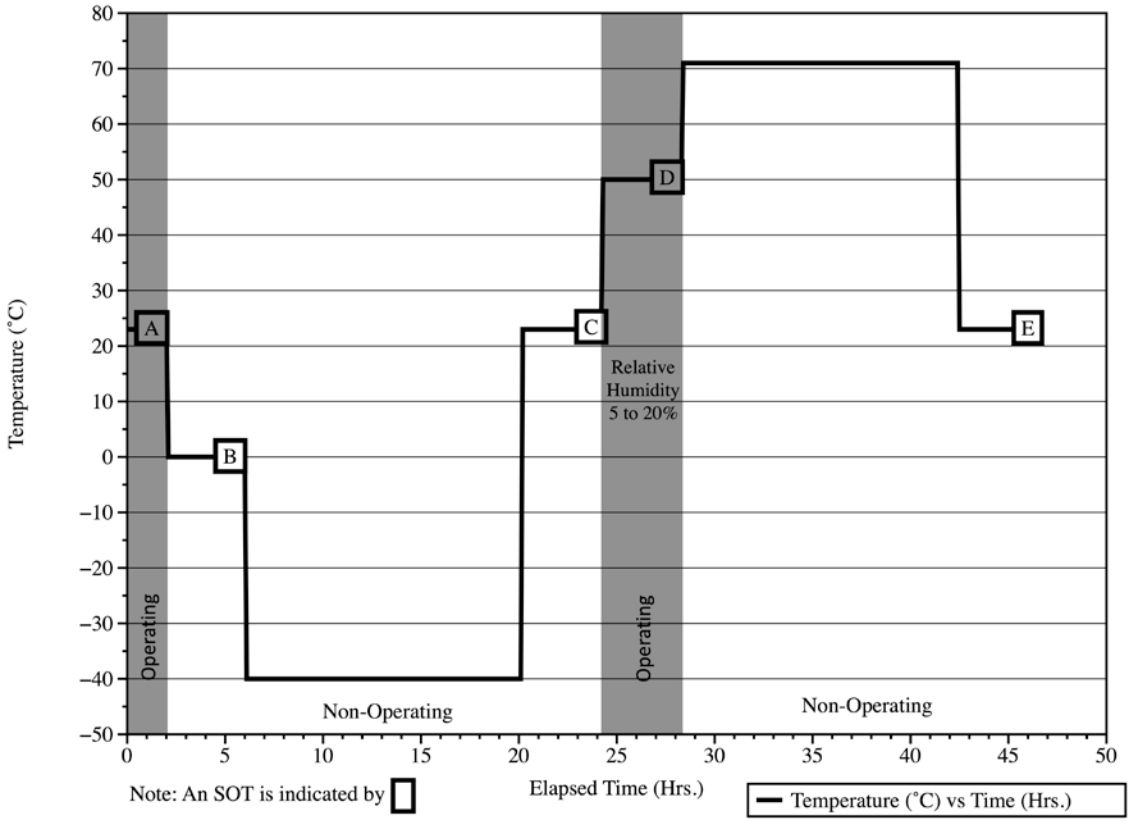


FIGURE 7. Class 3 sample temperature profile (including the arid climate test).

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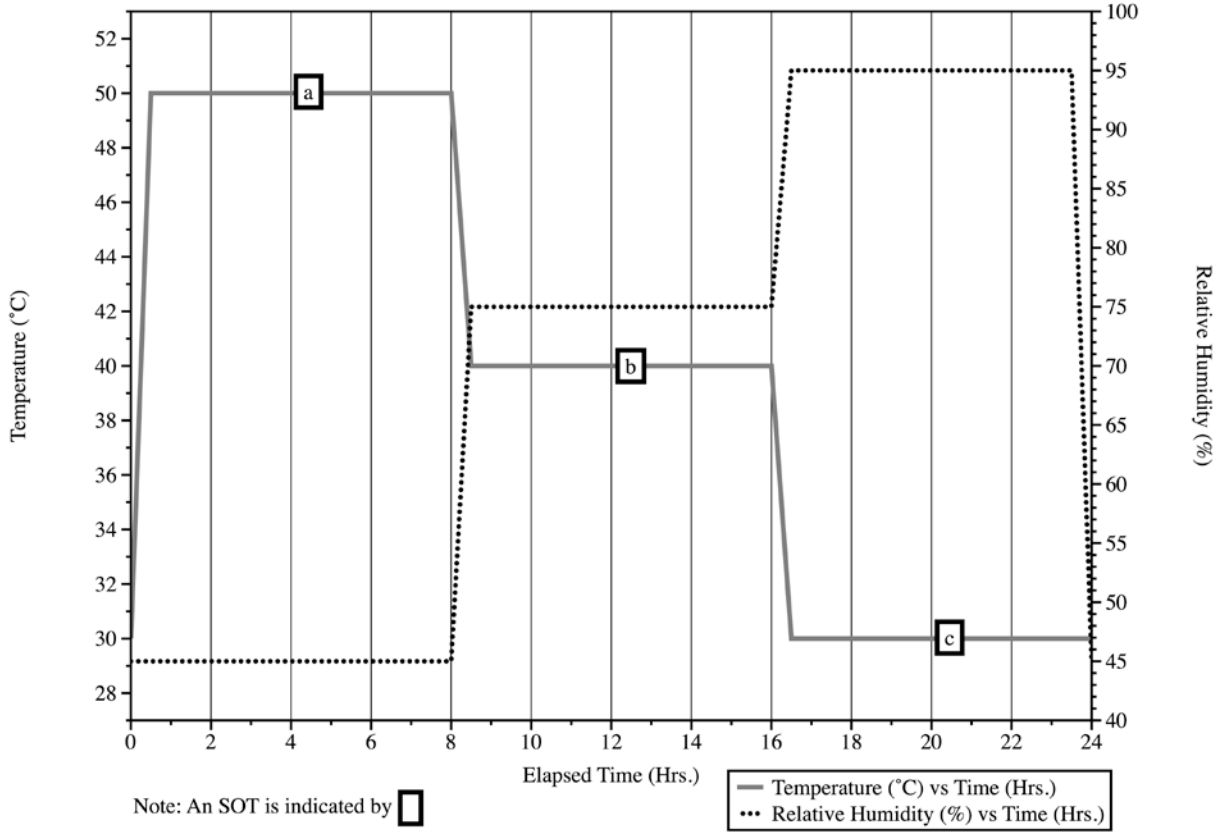


FIGURE 8. Class 3 sample humidity cycle profile.

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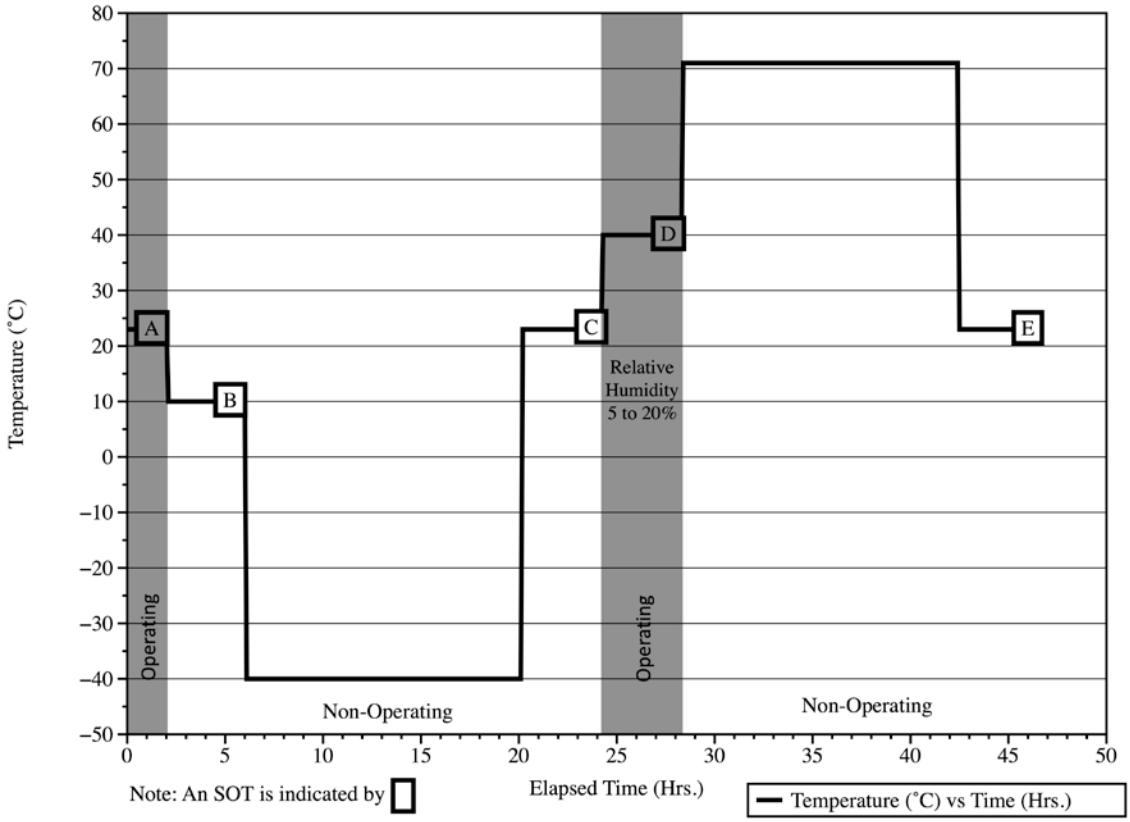


FIGURE 9. Class 4 sample temperature profile (including the arid climate test).

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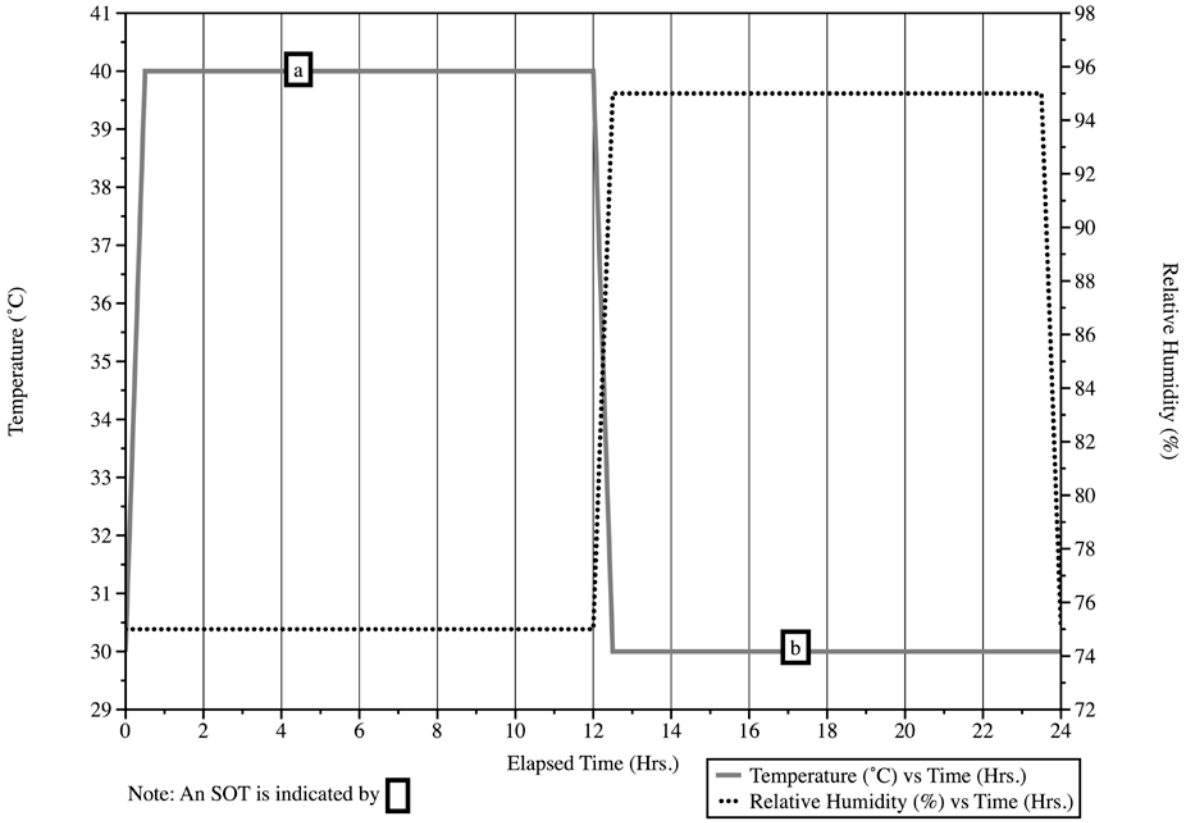


FIGURE 10. Class 4 sample humidity cycle profile.

4.5.5.2 Altitude test. The altitude test shall be performed as specified in table XIII for operating and nonoperating states at the simulated altitude. For the operating test, the following steps shall be performed with the test item operating at all times, allowing for proper warm-up time as specified in 3.9.1. For the nonoperating altitude test, the following steps shall be performed with the test item nonoperating except for the SOCs, allowing for proper warm-up time as specified in 3.9.1.

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TABLE XIII. Altitude test procedure.

Test step	Test procedure
Step 1	The test item shall be prepared in accordance with 4.5.2. The temperature shall be maintained within the specified operating range for the duration of the test. The test item shall be configured in a mode that can easily be accessed for measurement of conformance to specification requirements. An initial SOC shall be made at this point to determine conformance to specification requirements of the test item.
Step 2	The chamber pressure shall be decreased to simulate an altitude of 4600 meters [1/] at a rate not to exceed 600 meters per minute. This pressure shall be maintained for not less than 1 hour and then an SOC shall be performed (operating test only).
Step 3	The chamber shall be returned to standard ambient pressure at a rate not to exceed 600 meters per minute.
Step 4	The SOC shall be performed after return to the test conditions of 4.5.1.1. Degradation of equipment performance beyond the specified requirements shall constitute a failure.
NOTE: [1/] 3000 meters for class 4.	

4.5.5.3 Vibration tests. The vibration tests shall be as specified herein. If an equipment failure occurs, the unit under test may be repaired at the discretion of the procuring activity. If repair is allowed, testing shall continue from the point at which the failure occurred until the remaining test period is completed. The portion of the test period prior to the failure shall be repeated to evaluate the integrity of the repair. Failure modes that are not related to the original failure shall be disregarded during the retest. At the discretion of the procuring activity, a second unit may be subjected to the test in lieu of retesting the unit that failed.

4.5.5.3.1 Random vibration test.

4.5.5.3.1.1 Background information. Vibration levels of figure 11 shall be applied to class 1 equipment with durations of 1 hour per axis. Vibration levels of figure 12 shall be applied to class 2 equipment with durations of 30 minutes per axis. Vibration levels of figure 13 shall be applied to class 3 and 4 equipment with durations of 10 minutes per axis. The equipment shall be powered off during the vibration test. The equipment shall be hard mounted to the table by gripping the equipment's structure. Unless the equipment's feet are integral parts of the structure, they shall be removed during the test; if they are integral, the equipment shall be fixed so that the vibration is applied to the structural frame of the equipment. At the conclusion of the vibration test, a physical evaluation and a performance test of the equipment shall be conducted. These exposures are based on the typical size of equipment. When equipment is too large, unnecessarily high loads are induced in mounting and chassis structures, while higher frequency vibrations experienced at subassemblies are too low. In these cases, the test shall be applied to subassemblies. The maximum weight of an item or subassembly shall be approximately 36 kg.

4.5.5.3.1.2 Test level. The test levels shown on figures 11, 12, and 13 are for use in testing class 1, 2, and class 3 and 4, respectively.

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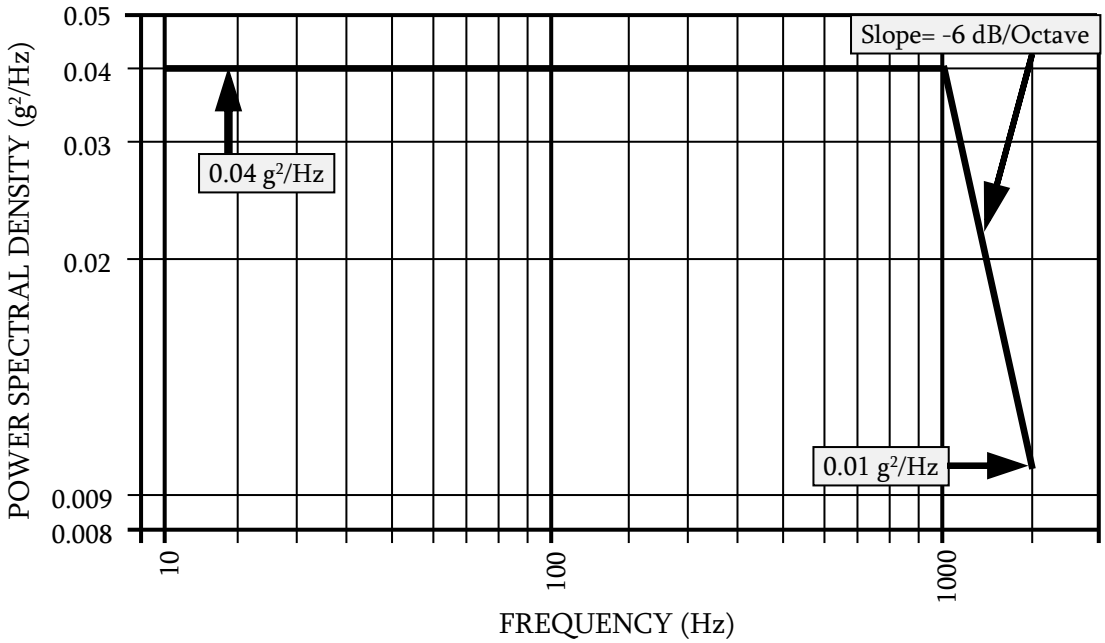


FIGURE 11. Vibration profile for class 1.

4.5.5.3.1.3 Test duration. The durations are 60 minutes per axis for class 1, 30 minutes per axis for class 2, and 10 minutes per axis for class 3 and 4.

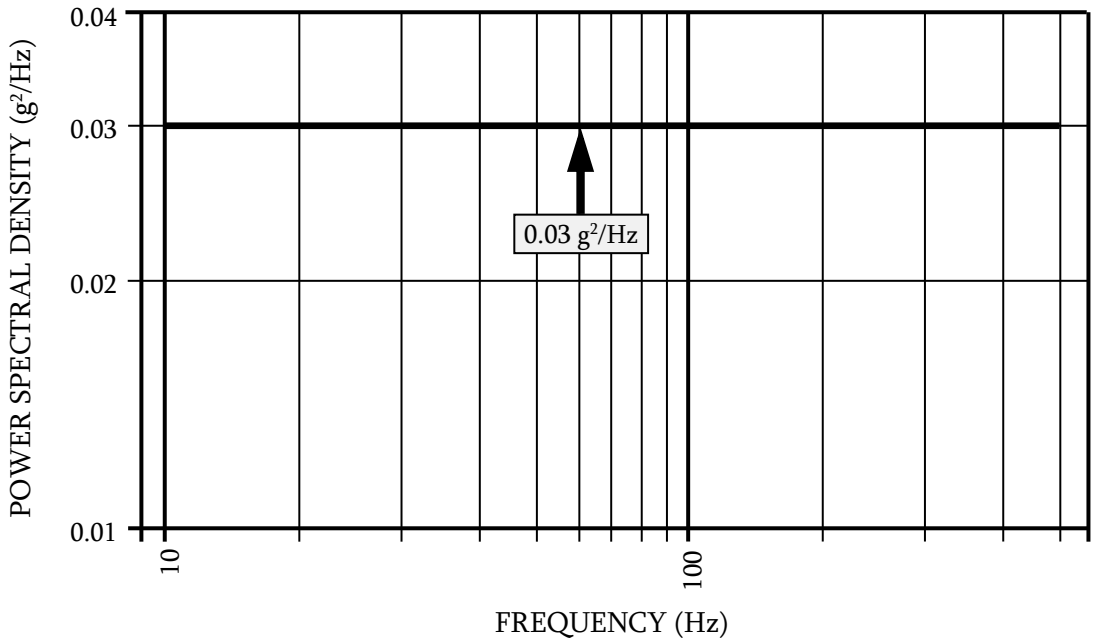


FIGURE 12. Vibration profile for class 2.

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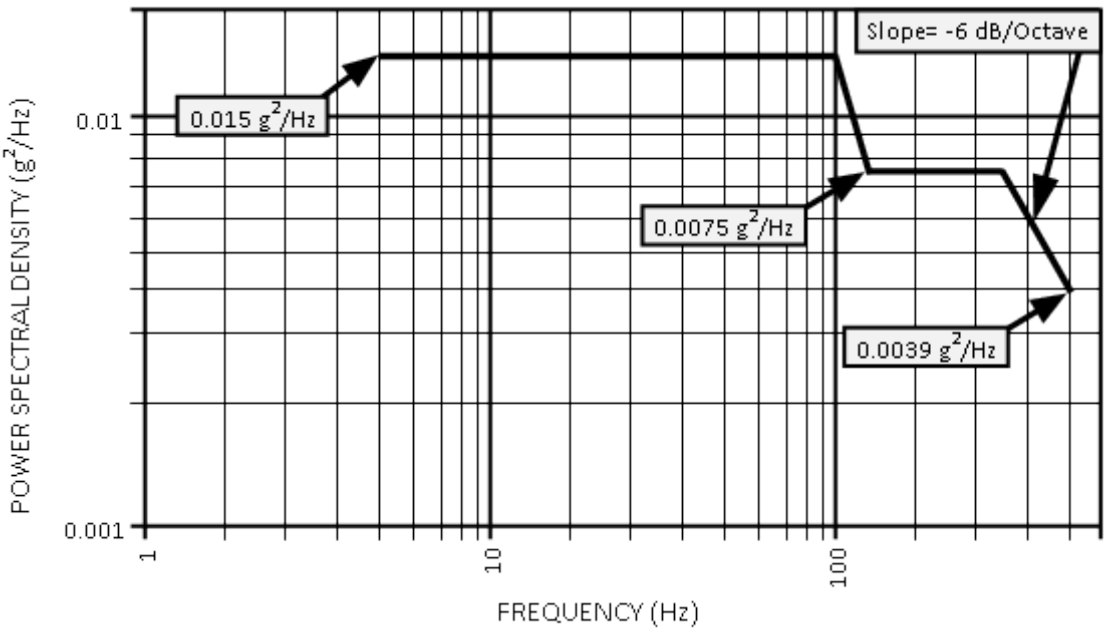


FIGURE 13. Vibration profile for class 3 and 4.

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4.5.5.3.2 Sinusoidal vibration test. When specified (see 6.2), sinusoidal vibration shall be performed on the subject class 1, 2, 3, and 4 equipment in lieu of random vibration testing of 4.5.5.3.1. Testing shall be in accordance with table XV and the procedures specified in table XIV.

TABLE XIV. Sinusoidal vibration test.

Test step	Test procedure
Step 1	The equipment shall be attached or strapped directly to the vibration table or to an intermediate structure designed to be capable of transmitting the specified magnitudes of vibration to the points of equipment attachment throughout the required test frequency range.
Step 2	During vibration, the equipment shall be operating as in the SOC mode.
Step 3	The frequency displacement and time shall be as specified in table XV. Resonance search, cycling, and dwell test times shall be continuous for the time periods specified. Testing shall not be performed below 10 Hz for class 3 and 4 equipment.
Step 4	A resonance search, cycling, and resonance dwell shall be conducted in each axis as follows: a. Resonance search. A resonance search shall be performed at vibration levels less than those specified in table XV but with sufficient amplitude to excite the test item. A difference of 6 dB or more between the excitation source and the test item or a part thereof shall indicate the presence of a resonance. If there is an audible or visual indication of resonance or any vibration-related malfunction of the equipment, an individual resonance test may be performed using the full vibratory test level and cycling time specified for the cycling test provided the resonance search time is included in the required cycling time. The test item or a subpart thereof is considered at resonance if the vibration amplitude is more than twice the amplitude of the vibration table. b. Cycling. The test item shall be vibrated in accordance with the applicable test levels, frequency range, and times specified in table XV. The frequency of applied vibration shall be swept over the specified range. The specified cycling time is that of an ascending plus a descending sweep and is twice the ascending sweep time. c. Resonance dwell. The test item shall be vibrated at the most severe resonant frequency determined in the resonance search for that axis. Test levels, frequency ranges, and test times shall be in accordance with table XV. If a change in the resonant frequency occurs during the test, the frequency shall be adjusted to maintain the peak resonance condition. If no significant resonant response is found, the equipment shall be vibrated for 10 minutes (resonance dwell) at 33 Hz.
Step 5	The SOTs shall be performed.

TABLE XV. Sinusoidal vibration test limits.

Class	Frequency increments (Hz)	Displacement (mm) peak-to-peak	Resonance search time per increment (minutes 1/)	Cycling time per axis (minutes)	Resonance dwell time per axis (minutes)
1 and 2	5 to 15 16 to 25 26 to 55	5 5 5	15	15	10
3 and 4	5 to 55	0.33	15	15	10

NOTE:
1/ Search times specified shall be for a low frequency to high frequency to low frequency sweep of the applicable increment.

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4.5.5.3.3 Loose cargo bounce test. All handheld equipment supplied with a hard transit case, and when other types of equipment are specified (see 6.2), shall be subjected to the loose cargo bounce test while contained in the hard transit case as specified in table XVI.

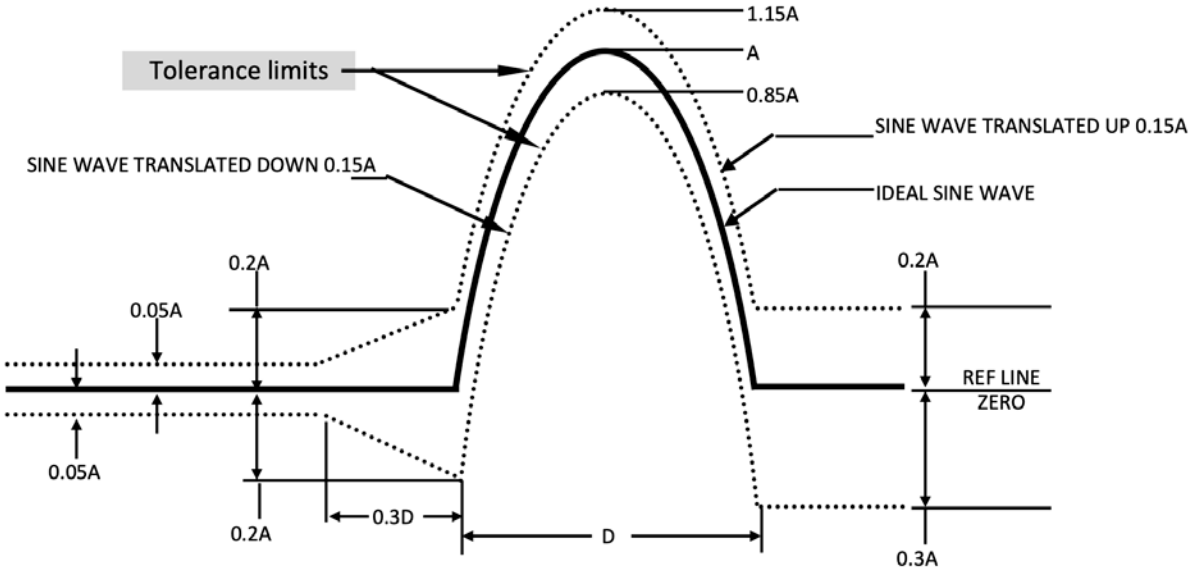
TABLE XVI. Loose cargo bounce test.

Test step	Test procedure
Step 1	The test apparatus shall be a package tester capable of 25.4 mm (double amplitude) displacement.
Step 2	The test bed of the package tester shall be covered with a panel of 12.7-mm (½-inch) plywood with the grain parallel to the drive chain. The plywood shall be secured with six-penny nails with the top of the heads flush with or slightly below the surface. Nails shall be spaced at 150-mm intervals around all four edges. If the distance between either pair of fences is greater than 600 mm, the plywood shall also be nailed at 75-mm intervals in a 150-mm square at the center of the test area. Using wooden fences, the test item shall be constrained to a horizontal motion of not more than 50 mm in a direction parallel to the axes of the shafts, which is a distance sufficient to ensure the test item will not rebound from fence to fence. For large items, care shall be taken to avoid potential air cushioning effects that may exist between the surface of the package tester and the test item.
Step 3	The test item, as secured in its transit case, equipment case, or as otherwise prepared for field transportation, shall be placed on the package tester within the constraints specified in step 2 above. The test item will not be operated during vibration. The package tester shall be operated in the synchronous mode with the shafts in phase. In this mode, any point on the bed of the package tester moves in a circular path in a vertical plane perpendicular to the axes of the shafts. The package tester shall be operated at 25.4-mm double amplitude and 284±2 rpm for a total of 3 hours. At the end of each half hour period, the test item shall be turned to rest on a different face, so that at the end of the 3-hour period, the test item will have rested on each of its six faces (top, bottom, sides, and ends). At the end of the 3-hour period, the test item shall be operated and inspected, and the results shall be obtained in accordance with 4.5.2.4. Minor surface abrasion shall not be cause for failure. The package tester shall be operated in the vertical linear mode (straight up and down in the vertical plane) instead of in the synchronous mode when one of the following conditions occurs: a. Bouncing of the test item is very severe and presents a hazard to personnel. b. Forward and rear oscillations cannot be reduced. When operated in the vertical linear mode, wooden fences shall be placed on all four sides of the test item to constrain the test item's motion to not more than 50 mm in either direction.

4.5.5.4 Mechanical shock tests.

4.5.5.4.1 Functional shock test. Equipment shall be subjected to functional shock testing. A half-sine shock pulse with the configuration and tolerance limits shown on figure 14 shall be used. The duration (D) of the pulse shall be 11 ms, and the peak shock value (A) shall be 30g. The equipment shall operate during the test. The equipment shall be given three shocks in each direction and on each axis for a total of 18 shocks. The SOT shall be performed after shock exposure and a physical evaluation shall be conducted.

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NOTE: The oscillogram shall include a time about 3D long with the pulse approximately in the center. The acceleration amplitude of the ideal half-sine pulse is A and its duration is D. The measured acceleration pulse shall be contained between the broken line boundaries and the measured velocity change, which may be obtained by integration of the acceleration pulse, shall be within the limits Vi±0.1 Vi, where Vi is the velocity-change associated with the ideal pulse that equals 2 AD/p. The integration to determine velocity change shall extend from 0.4D before the pulse to 0.1D after the pulse.

FIGURE 14. Half-sine shock pulse configuration and limits.

4.5.5.4.2 Transit drop test. Equipment for which the enclosure acts as the transit case, equipment that is specified to include a transit case (see 6.2), and all class 1 equipment shall be subjected to the transit drop test. The drop height and number of impacts shall be as specified in table XVII. For equipment 454 kg or less, the floor or barrier receiving the impact shall be of 50-mm plywood backed by concrete. For equipment over 454 kg, the floor or barrier shall be concrete.

TABLE XVII. Transit drop test.

Equipment class	Drop height	Number of impacts 1/
Class 1	460 mm	10
All classes where the enclosure is intended to act as the transit case	300 mm	10
Class 3 and 4 where the enclosure is intended to act as the transit case	200 mm	10
All classes, when an airtight transit case is specified 2/	460 mm	10
All classes, when a hard transit case is specified 2/	460 mm	10
All classes, when a soft transit case is specified 2/, 3/	610 mm	2 per strap

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TABLE XVII. Transit drop test – Continued.

NOTES:
1/ The equipment shall be dropped on the four bottom corners and six faces for the total number of impacts divided between the test items.
2/ Equipment shall be contained in a transit case throughout the test.
3/ Only those cases with shoulder straps shall be subjected to this test. The shoulder strap shall be supported at a point sufficient to allow a fall of 600 mm while preventing the enclosure from contacting the impact surface. The equipment shall remain inside the transit case throughout this test.

4.5.5.4.3 Bench handling test. With the equipment operating as in the SOC mode, the equipment shall be placed on a horizontal, solid wood bench top at least 4.1 cm thick in a position that allows for servicing. The bench handling test shall be performed as specified in table XVIII in a manner simulating shocks liable to occur during servicing.

TABLE XVIII. Bench handling test.

Test step	Test procedure
Step 1	Using one edge as a pivot, the opposite edge of the chassis shall be lifted until one of the following conditions occurs (whichever occurs first): a. The chassis forms an angle of 45 degrees with the horizontal bench top. b. The lifted edge of the chassis has been raised 100 mm above the horizontal bench top. c. The lifted edge of the chassis is just below the point of perfect balance. The chassis shall be allowed to drop back freely to the horizontal bench top. A total of four drops shall be repeated using other practical edges of the same horizontal face as pivot points.
Step 2	Step 1 shall be repeated with the test item resting on other faces until the test item has been dropped for a total of four times on each face on which the test item could reasonably be placed during its servicing.
Step 3	Steps 1 and 2 shall be repeated with the test item not operating and the cabinet or case removed except for equipment where the case serves as the only chassis or support structure.
Step 4	The test item shall be examined for mechanical damage. Damage to the equipment, other than cosmetic, shall constitute failure.
Step 5	The SOT shall be performed.

4.5.5.4.4 High-impact shock test. The high-impact shock test shall be as specified (see 6.2).

4.5.5.5 Water resistance tests. When specified (see 6.2), the water resistance tests specified herein shall be performed after the required mechanical shock tests.

4.5.5.5.1 Class 1 and airtight transit case watertight test. With any case cover latched in place and all drains and ports closed, class 1 equipment and equipment in an airtight transit case with the equipment installed shall be subjected to a submergence test for watertight equipment. The assembly shall be submerged such that there is a minimum of 0.9 meter of water above the top of the enclosure. The equipment shall be not operating while the cover is latched in place. The SOT shall be performed after submergence.

4.5.5.5.2 Class 1 splash-proof test. With the cover off and the equipment operating, the steps specified in table XIX shall be performed.

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4.5.5.5.3 Class 2, 3, and 4 and hard transit case drip-proof test. When class 2, 3, or 4 equipment is specified as drip-proof, the drip-proof test shall be performed. When a hard transit case is specified, the drip-proof test shall be performed only when the test is specified. Unless otherwise specified (see 6.2), the equipment shall not operate during the test. Unless otherwise specified (see 6.2), equipment covers (if present) or the transit case shall be closed. The drip-proof test shall be performed as specified in table XIX.

TABLE XIX. Drip-proof test.

Test step	Test procedure
Step 1	The test item (equipment or equipment in transit case) shall be placed in the chamber in its normal operating position on a platform or tilt table in a manner such that it can be oriented horizontally and in any of four directions forward, backward, and to each side to the extent of 15 degrees from the horizontal plane.
Step 2	A spray nozzle with a uniformly distributed full cone spray pattern shall be attached to a flexible hose and, using a container calibrated in liters or an equivalent method, the rate of flow out of the nozzle shall be determined. The requirements for water flow shall be 175 liters per hour for each square meter of area covered by the spray 0.9 meter below the nozzle.
Step 3	The nozzle shall be positioned 0.9 meter above the test item so that the spray is directed downward onto the test item. The test item shall be subjected to a uniform spray over the entire surface being tested. Unless otherwise specified (see 6.2), equipment shall not operate during the test.
Step 4	The spray shall be directed onto the top of the test item (with platform or tilt table horizontal) for at least 8 minutes. The spray shall be turned off for 4 minutes.
Step 5	The spray shall be directed downward onto the test item for at least 8 minutes with the platform or table tilted such that the test item is oriented 15 degrees from the horizontal in the forward position. The spray shall be turned off for 4 minutes
Step 6	The spray shall be directed downward onto the test item for at least 8 minutes with the platform or table tilted such that the test item is oriented 15 degrees from the horizontal in the backward position. The spray shall be turned off for 4 minutes.
Step 7	The spray shall be directed downward onto each remaining side of the test item for at least 8 minutes with the platform or table tilted such that the side being sprayed is oriented 15 degrees from the vertical. The spray shall be turned off for 4 minutes.
Step 8	The test item shall be returned to the horizontal (platform or tilt table horizontal) and the SOT shall be performed.

4.5.5.6 Electrical power tests. During the electrical power tests, the equipment performance shall be monitored as specified herein.

4.5.5.6.1 Input power test. The input power shall be tested by measuring the power consumption of the equipment. Power consumption that exceeds the value specified (see 6.2) shall constitute a failure of this test.

4.5.5.6.2 Voltage and frequency variation test. The response of the equipment to input voltage and frequency variations shall be tested as specified in table XX.

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TABLE XX. Voltage and frequency variation test.

Test step	Test procedure
Step 1	The power cable of the equipment shall be connected to a variable voltage and frequency source.
Step 2	The output of the variable source shall be set at 120 Vrms, 60 Hz.
Step 3	Performance of the equipment outside the limits required in the SOC or other tests specified (see 6.2) shall constitute a failure of this test.
Step 4	Steps 2 and 3 shall be repeated for voltages and frequencies as specified (see 6.2) from the variable source as specified in table XXII using the maximum voltage and minimum voltage limit values. The voltage and frequency shall be maintained for at least 15 minutes at each level specified. This test shall be performed at other points within the specified ranges at the discretion of the Government.

4.5.5.6.3 Voltage-transient test. The equipment performance during voltage transients shall be tested as specified in table XXI.

TABLE XXI. Voltage-transient test.

Test step	Test procedure
Step 1	The power cable of the equipment shall be connected to a 60-Hz variable voltage source.
Step 2	The output of the source shall be set to the maximum voltage for the equipment type as specified in table XXII.
Step 3	The source voltage shall be quickly increased to a transient value of 156 Vrms, held for approximately 0.5 second, and quickly returned to the starting voltage of step 2. Equipment performance shall be monitored during and after the transient for compliance with the SOC or other tests specified (see 6.2). Performance outside of the specified limits for longer than 30 seconds shall constitute a failure of this test.
Step 4	Step 3 shall be repeated four more times at 2-minute intervals.
Step 5	The output of the source shall be set to a starting voltage of 108 Vrms.
Step 6	Step 3 shall be repeated with a transient value of 84 Vrms and the starting voltage specified in step 5.
Step 7	Step 6 shall be repeated four more times at 2-minute intervals.

TABLE XXII. Variable frequency power supply outputs.

Frequency (Hz)	Minimum voltage (Vrms)	Maximum voltage (Vrms)
47.5	108	132
52.5	108	132
57	108	132
63	108	132
380	108	132
420	108	132

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4.5.5.6.4 Frequency-transient test. The equipment performance during frequency transients shall be tested as specified in table XXIII.

TABLE XXIII. Frequency-transient test.

Test step	Test procedure
Step 1	The power cable of the equipment shall be connected to a 120-Vrms variable frequency source. The output voltage shall be the operating voltage specified for the equipment.
Step 2	The output of the source shall be set to a starting frequency of 47.5 Hz.
Step 3	The source frequency shall be quickly changed to a transient value of 45 Hz, held for approximately 0.5 second, and quickly returned to the starting frequency of 47.5 Hz. Equipment performance shall be monitored during and after the transient for compliance to the SOC or other tests specified (see 6.2). Performance outside of the specified limits for longer than 30 seconds shall constitute a failure of this test.
Step 4	Step 3 shall be repeated four more times at 1-minute intervals.
Step 5	The output frequency of the source shall be set to a starting frequency of 57 Hz.
Step 6	Step 3 shall be repeated with a transient frequency of 54 Hz and a starting frequency of 57 Hz.
Step 7	Step 6 shall be repeated four more times at 1-minute intervals.
Step 8	The output frequency of the source shall be set to a starting frequency of 63 Hz.
Step 9	Step 3 shall be repeated with a transient frequency of 66 Hz and a starting frequency of 63 Hz.
Step 10	Step 9 shall be repeated four more times at 1-minute intervals.
Step 11	The output frequency of the source shall be set to a starting frequency of 380 Hz.
Step 12	Step 3 shall be repeated with a transient frequency of 360 Hz and a starting frequency of 380 Hz.
Step 13	Step 12 shall be repeated four more times at 1-minute intervals.
Step 14	The output frequency of the source shall be set to a starting frequency of 420 Hz.
Step 15	Step 3 shall be repeated with a transient frequency of 440 Hz and a starting frequency of 420 Hz.
Step 16	Step 15 shall be repeated four more times at 1-minute intervals.

4.5.5.6.5 Power source interruption test. The power source interruption test shall conform to the requirements of 3.6.1.2.3.

4.5.5.7 Group C verification test. The group C verification test shall be performed after the required tests of 4.5.5.1 through 4.5.5.6.5 have been completed. The group C verification test shall consist of the group A and group B tests as specified by class and 6.2. Group A and group B test failure criteria shall apply.

4.5.6 Group D tests. Group D tests shall be performed as specified herein and in the approved equipment test procedure. These tests shall be performed under the test conditions specified in 4.5.1.1.

4.5.6.1 Fungus resistance test. When specified (see 6.2), the equipment shall be subjected to method 508 (fungus) as specified in MIL-STD-810. The five species of test fungus specified for method 508 shall be used for the test. The test duration shall be 28 days with the equipment in a non-operational state. Pre- and post-operation tests using the SOC are required.

4.5.6.2 Salt fog tests. The equipment shall be subjected to method 509 (salt fog) as specified in MIL-STD-810. The test duration shall be 48 hours with the samples being constantly wetted with a 5±1 percent salt solution. A 48-hour drying period in a standard ambient atmosphere shall follow.

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4.5.6.2.1 Enclosure (salt fog test). When specified for any class or transit cases (see 6.2), the equipment or transit case (with equipment enclosed) shall be subjected to the salt fog test. Class 1 equipment shall have the cover (if a cover is present) removed from the enclosure and placed in the salt fog chamber with the accessories (except the Use and Installation Manual and those accessories used in the SOT) packed in the normal manner in the cover. All classes of equipment (if a transit case is specified [see 6.2], the cables and connectors shall be removed from the equipment in the transit case before the transit case is closed) shall be not operating but with cables and connectors attached as for the SOT. Unused connectors may have caps or covers installed. The SOT specified (see 6.2) shall be performed before any cleaning is performed. The equipment shall be inspected to detect evidence of physical degradation. Visible corrosion is permissible if it is removed by cleaning and the equipment is returned to a condition that conforms to the requirements of 4.5.3.1. The SOT shall be performed as specified (see 6.2). Degradation of equipment performance beyond the requirements of this specification shall constitute a failure of this test.

4.5.6.2.2 Structural parts (salt fog test). Unless otherwise specified for class 1 equipment (see 6.2), and when specified for other classes of equipment (see 6.2), the equipment shall have one or more samples of each material used for the fabrication of panels and structural parts (chassis, rails, braces, etc.) with finishes as in the completed equipment subjected to the salt fog test. After exposure, the material shall be inspected to detect any evidence of physical degradation. Visible corrosion is permissible if it is removed by cleaning and the material is returned to a condition that conforms to the requirements of 4.5.3.1. Degradation of material beyond the requirements of the intended use shall constitute a failure of this test.

4.5.6.2.3 Procedure. The equipment shall be in its normal operating mode (except for equipment in transit cases) without power applied. The equipment (or equipment in its transit case) shall be stored in a standard ambient atmosphere for 48 hours following the 48-hour salt exposure. At the end of the drying period, SOTs shall be performed to verify that equipment performance has not deteriorated.

4.5.6.3 Explosive atmosphere test. Class 1 equipment (and any equipment of a different class as specified in 6.2) shall be subjected to procedure I of method 511 (explosive atmosphere) as specified in MIL-STD-810. The maximum operating parameters for temperature, relative humidity, and altitude as specified in table II shall be used as applicable for the class of equipment being tested.

4.5.6.4 Dust resistance test (class 1 and hard transit cases). Class 1 equipment (and any equipment of a different class as specified in 6.2) and equipment in hard transit cases shall be subjected to procedure I of method 510 (sand and dust) as specified in MIL-STD-810. Class 1 equipment shall be tested with the equipment not operating and with the cover on. Equipment in transit cases shall have the transit case closed and secured. The test shall be conducted with small particle silicone flour at an air velocity of 8.9 m/s. The SOT shall be performed before and after the dust test. Degradation of equipment performance beyond the requirements of this specification shall constitute a failure of this test. The test item shall have a minimum clearance of 150 mm from any wall of the test chamber and from any other test item (if more than one item is being tested). The equipment shall be operated in accordance with the SOC before, during, and after the dust test.

4.5.6.5 EMC test. The equipment shall be subjected to the EMC tests and limits specified in 3.10. The measuring equipment, set up, procedures, and operation of the equipment shall be in accordance with IEC CISPR 11, class A; IEC 61000-4-2; IEC 61000-4-3; IEC 61000-4-4; and IEC 61000-4-6. Test conditions shall be in accordance with 4.5.1.1.

4.5.6.6 Magnetic environment test. When specified (see 6.2), the magnetic environment test specified in table XXVI shall be performed. The magnetic environment test shall be conducted with a Helmholtz coil set that is 1 meter in diameter, each coil not over 0.13 meter in breadth, spaced 0.5 meter apart, and capable of carrying sufficient current to produce a field intensity of 20 oersteds. Larger coil diameters and separation may be used when the dimensional ratios are as specified herein.

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TABLE XXVI. Magnetic environment test.

Test step	Test procedure
Step 1	Prior to placing equipment in the area of the field, the coil set shall be energized with a DC current to produce a 20±1 oersted field in air. The magnitude of current required shall be observed and the coil set shall be de-energized.
Step 2	The equipment shall be placed in the center of the coil set.
Step 3	The equipment shall be operated. The performance of the equipment shall be verified in accordance with the SOT.
Step 4	The coil set shall be energized so that the equipment is subjected to a magnetic field that is increased from 0 oersteds to 20±1 oersteds at a rate of change between 20 and 200 oersteds per second (using the current observed in step 1). The field shall be maintained at 20±1 oersteds for 5 minutes or longer, if necessary, to verify correct operation in the SOC. The equipment shall be observed for any maloperation. If an effect is observed, the magnitude of the field shall be reduced at a rate of change between 20 oersteds and 200 oersteds per second until the effect is no longer apparent and the magnitude of the field shall be observed at that point. The coil set shall be de-energized. Reduction of the field to less than 5 oersteds to permit specified operation of the equipment shall constitute a failure of this test.
Step 5	The connections of the coil to the power source shall be reversed and steps 4 and 5 shall be repeated.
Step 6	Steps 3 through 5 shall be performed for the three major axes of the equipment.
Step 7	The SOT shall be performed to verify satisfactory operation of the equipment.

4.5.6.7 Group D verification tests. The group D verification tests shall be performed after the required tests of 4.5.6.1 through 4.5.6.6 have been completed. This test shall consist of the group A and group B tests specified by class and in 6.2. Group A and B test failure criteria shall apply.

4.5.6.8 Solar radiation test. When specified (see 6.2), class 1 and 2 equipment shall be subjected to procedure I of method 505 (solar radiation) as specified in MIL-STD-810. The temperature category shall be class A2 as specified for method 505 in MIL-STD-810. Full spectrum lamps shall be used. The item shall be orientated to expose its most vulnerable parts to the solar radiation, unless a prescribed orientation sequence is specified in the test plan (see 6.6.29). The equipment shall be operated in accordance with the SOC before, during, and after the solar radiation test.

TABLE XXVIII. Temperature/solar radiation diurnal cycle.

Time (hours)	Temperature (°C)	Solar radiation (W/m^2)
0000	33	0
0300	32	0
0600	30	55
0900	37	730
1200	42	1,120
1500	43	915
1600	43	730
1800	42	270
2100	36	0
2400	33	0

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4.5.7 Group E tests. Group E tests shall be performed as specified herein and in the approved equipment test procedure.

4.5.7.1 Maintainability tests.

4.5.7.1.1 Maintainability demonstration. Equipment shall be evaluated for conformance to the requirements specified in 3.15.1 through 3.15.4.

4.5.7.1.2 Washability test. When specified (see 6.2), equipment shall be subjected to the washability test specified in table XXIX. The washability test shall be performed prior to the maintainability demonstration.

TABLE XXIX. Washability test.

Test step	Test procedure
Step 1	The equipment case or covers shall be removed to expose internal parts and circuits as for servicing.
Step 2	Using a hand-atomizer or paint-sprayer, a thin coat of light machine oil containing a coloring agent such as food coloring shall be sprayed over the exposed interior and exterior parts. The coating shall be sufficient to cause easily recognizable discoloration.
Step 3	The equipment shall be subjected to a 50 °C temperature for 24 hours with suitable precautions against accumulation of flammable vapor.
Step 4	The equipment shall be subjected to water-wash according to the equipment manufacturer's washing procedure and with detergents or cleaners specified for the equipment by the manufacturer, until the added contaminants of step 2 have been removed. The equipment shall be rinsed and dried as specified in the manufacturer's procedures.
Step 5	Relubrication shall be performed as specified in the manufacturer's procedures.
Step 6	The equipment shall be inspected. Loss of finishes or markings, deformation of parts, or other evidence of deterioration shall constitute a failure of this test.
Step 7	The equipment shall be operated with particular attention to the ease of operation of controls and switches and functional stability. The equipment shall be operated as in the SOT. Degradation of equipment performance beyond the specified requirement shall constitute a failure of this test.

4.5.7.2 Acoustic noise test. Equipment shall be tested in accordance with ISO 7779 to ensure conformance to the requirement specified in 3.9.12.

4.5.7.3 Dimensions and weight tests. Compliance with the specified weight and dimensional requirements of the equipment, and any plug-ins or accessories for which the dimension and weight are separately specified (see 6.2), shall be determined by weighing and measuring the equipment.

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4.5.7.4 Mechanical stability test. Unless excluded in 3.7.3, equipment shall be subjected to the mechanical stability test specified table XXX.

TABLE XXX. Mechanical stability test.

Test step	Test procedure
Step 1	The test item shall be positioned on a raised platform or table with a surface area of sufficient size to accommodate the total bottom surface of the test item in its normal operating position and configuration.
Step 2	The test item shall be repositioned so that ¼ of the bottom surface extends beyond the edge of the platform or table. The extended side of the test item shall be parallel to the edge of the platform that is located under the test item.
Step 3	The test item shall be observed for evidence of instability. Evidence of instability shall be considered a test failure.
Step 4	Steps 2 and 3 shall be repeated for all additional vertical sides of the test item.
Step 5	Steps 2 through 4 shall be repeated for each required alternate operating position and configuration of the test item. When the rear of the test item is equipped with feet or bumpers (such as for use on a floor or deck) that prohibit the repositioning of the test item in accordance with step 2, an auxiliary support may be provided for this step. The auxiliary support (such as a piece of plywood) shall be not larger than the bottom surface in its alternate position but shall be sufficiently large to accommodate the rear feet or bumpers. During the test, the support and the test item shall be repositioned as specified in step 2.

4.5.7.5 Equipment emanations test. When specified (see 6.2), the emanations tests for ionizing radiation, microwave radiation, liberated gases, and ultrasonic pressure shall be performed in accordance with IEC 61010-1.

4.5.7.6 Additional tests. The test and accept/reject limits for any group E additional tests shall be as specified (see 6.2).

4.5.8 Group F reliability tests. The reliability tests, when specified (see 6.2), shall consist of evaluating the mean time between failure (MTBF) based on actual field failure data and the analysis of that data as specified (see 6.2). Otherwise (i.e., newly introduced equipment for which there is insufficient data), data shall be in analysis form based on the closest similar actual product data as specified (see 6.2). Records of the actual equipment model shall be used unless the Government authorizes substitution of records for a similar model due to the lack of sufficient supporting documentation of the offered equipment. The analysis shall be able to reasonably demonstrate that the offered equipment provides reliability similar to the model used to provide substitute records.

4.6 Test plan, procedures, reports, and other test data. The manufacturer's test plan (see 6.6.29) shall be approved by the Government prior to the start of testing. The manufacturer's test plan, test procedures, test reports (see 6.6.31), and other test data shall be as specified (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Test equipment covered by this specification is intended for use in testing equipment and systems in the worldwide natural and controlled environments in which military equipment is operated (on a continuous or intermittent basis for extended periods of time), either stored or transported, or both, including utilization in all types of moving military and commercial vehicles (land, sea, and air). The extent of an equipment's penetration (levels of use) into the military environment is limited by its classification. Selection of the correct class is essential to ensure procurement of the desired equipment. Where needs supersede requirements, this specification can be referenced in part. Where environmental needs overlap specific classes, the highest appropriate class (lowest environmental resistance) will form the basic requirements, and the additional specific requirements will be selected from the lower classes. The inclusion of batteries may alter the environmental resistance of equipment designed to operate from either internal or external power sources or both. Any existing general purpose electronic/electrical test equipment assets or stock that have been procured to MIL-PRF-28800F, dated 24 June 1996, or to earlier versions are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Detailed requirements for particular test equipment (see 1.1).

c. Appropriate class of test equipment (see 1.2).

d. When a bid sample is required (see 3.1.1, 4.2, 6.3, and A.2.1), the number of bid samples to be submitted if other than specified (see 4.2), and return shipping instructions of the equipment provided for bid sample inspection (see A.2.2.e).

e. When first article inspection is required (see 3.1.2 and 6.4), the number of first article samples to be submitted if other than specified (see 4.3), when group B testing is not required (see 4.3.b), and when additional group E tests are required (see 4.3.5.f).

f. Finish required, if other than what is normally provided by the manufacturer (see 3.2.1).

g. Color required, if other than what is normally provided by the manufacturer (see 3.2.2).

h. Requirement for restricted materials and parts, if other than specified (see 3.2.4.1).

i. Cybersecurity requirements in accordance with local organizational policy (see 3.4).

j. When a digital control interface is required, the type of digital control interface required (IEEE 488, IEEE 802.3, or USB), and the measurement information to be transferred by the digital control interface (see 3.5, 3.5.1, 3.5.2, and 3.5.3).

k. Maximum power for operation (see 3.6.1.1).

l. When equipment is required to operate from self-contained batteries and the battery operating and recharge time requirements (see 3.6.1.2.5 and 3.6.2.7).

m. When maximum use of standard, nomenclatured battery-power sources is not required (see 3.6.1.2.5.1.a).

n. When the use of mercury or lithium batteries is authorized and the information required (see 3.6.1.2.5.1.b and 3.6.1.2.5.1.c).

o. External portable battery pack requirements, when required (see 3.6.1.2.5.4).

p. Requirements for equipment operation from an external DC source (see 3.6.1.2.6).

q. Requirements for power connectors and connections for equipment rated above 10A rms (see 3.6.2).

r. Requirements for equipment requiring multiphase power (see 3.6.2.1).

s. Input power cable required, if other than the detachable power cable normally provided by the manufacturer (see 3.6.2.2).

t. When equipment is required to provide an internal battery charger capable of charging the internal batteries when the equipment is connected to the specified AC power source. Requirements for battery charge capability (period of hours) and recharging time, if other than specified (see 3.6.2.8).

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u. When equipment powered from internal batteries is required to include an externally accessible, covered battery compartment (see 3.6.2.9).

v. If a COTS enclosure is not acceptable (see 3.7).

w. Enclosure dimensions (see 3.7.1) and maximum weight (see 3.7.2).

x. When the enclosure is required to have a cover (see 3.7.4.1).

y. Maximum stacking height of equipment enclosures (see 3.7.4.3) and transit cases (see 3.8.2).

z. Number of enclosure handles required, if other than specified (see 3.7.4.5).

aa. When the enclosure is required to stow all accessories (see 3.7.4.6).

bb. Location of external electrical connections and operating controls, if other than specified (see 3.7.4.8).

cc. Location of chassis, if other than specified (see 3.7.4.9).

dd. When a mechanism for equalizing air pressure within the enclosure is not required for class 1 equipment (see 3.7.4.10).

ee. When the equipment is required to be fitted with a console cabinet and additional requirements for transportability (see 3.7.4.11).

ff. When equipment fitted with a console cabinet is required to have work surfaces for personnel provided (see 3.7.4.12).

gg. When the equipment is required to be capable of being rack-mounted (see 3.7.4.13.2).

hh. When a soft or hard transit case is required (see 3.8), whether the transit case needs to be airtight (see 3.9.6.1.3), whether the soft transit case needs to be drip-proof (see 3.9.6.3.3), when the identification plate on the transit case needs to also contain the information specified in 3.12.2.1 (see 3.12.3), and the location of the identification plate on the transit case (see 3.12.3).

ii. Combinations of environmental conditions beyond those covered by the specified test procedures in 4.5 (see 3.9.1).

jj. Test procedure warm-up requirements, if other than specified (see 3.9.1).

kk. Environmental performance requirements (see table II).

ll. When class 1, 2, 3, and 4 equipment is required to conform to the specified performance and accuracy requirements when subjected to the sinusoidal vibration conditions specified in table IV in accordance with 4.5.5.3.2 (see 3.9.4.2).

mm. When other than class 1 and 2 equipment is required to conform to the loose cargo bounce test (see 3.9.4.3).

nn. Requirements for high-impact shock testing (see 3.9.5.4 and 4.5.5.4.4).

oo. When class 2, 3, or 4 equipment is required to be watertight (see 3.9.6.1.2), splash-proof (see 3.9.6.2.2), or drip-proof (see 3.9.6.3.1).

pp. When samples of class 1 panels and structural parts do not need to be tested for the salt atmosphere requirement (see 3.9.8.2).

qq. When equipment is for Navy use and the DC field magnetic environment test is applicable (see 3.9.11).

rr. Acoustic noise requirement, if other than specified (see 3.9.12).

ss. When class 1 and 2 equipment is required to conform to applicable requirements after exposure to solar radiation when tested in accordance with 4.5.6.8 (see 3.9.13).

tt. When a reusable pouch or container is required to be provided (see 3.12.4).

uu. Requirement for nomenclature assignment and the nomenclature to be assigned (see 3.12.5).

vv. Minimum equipment reliability requirements, if other than specified (see 3.14).

ww. Quantity of maintenance and calibration aid sets if required, including circuit board extenders, special adapters, special tools, and patch cables (see 3.15.3.b).

xx. When equipment is required to be subjected to the washability test (see 3.15.4).

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yy. When class 1 equipment is required to conform to environmental stress screening requirements (see 3.16).

zz. Requirement for the manufacturer's Use and Installation Manual and Maintenance and Service Manual to be provided when needed (see 3.17).

aaa. When conformance inspection group B testing is not required (see 4.4.b) and when additional group E tests are required (see 4.4.5.f).

bbb. Parameters and limits of acceptability for an SOC and the parameters for an SOT (see 4.5).

ccc. Additional or different failure criteria, if other than specified (see 4.5.2.5).

ddd. When water resistance testing is required (see 4.5.5.5).

eee. Whether equipment is required to operate during the drip-proof test and whether equipment covers (if present) or the transit case are not required to be closed during the drip-proof test (see 4.5.5.5.3).

fff. When fungus resistance testing is required (see 4.5.6.1).

ggg. When the salt fog test is required (see 4.5.6.2).

hhh. When the explosive atmosphere test is required for other than class 1 equipment (see 4.5.6.3).

iii. When the dust resistance test is required for other than class 1 equipment (see 4.5.6.4).

jjj. When group E emanations tests are required (see 4.5.7.5).

kkk. When group F reliability analysis is required (see 4.5.8). Requirements for MTBF field failure data and the analysis of that data or data in analysis form based on the closest similar actual product data (see 4.5.8).

lll. Requirements for test plans, test procedures, test reports, and other test data when needed (see 4.6).

mmm. Packaging requirements (see 5.1).

nnn. When bid sample inspection is required (see A.1.1).

6.3 Bid sample inspection. Bid sample inspection is performed by the Government before contract award. When a bid sample is required (see 6.2), the equipment should be a production unit normally provided by the manufacturer. The contracting officer should include specific instructions in the acquisition documents regarding arrangements for examinations and disposition of the bid samples. Invitations for bids should provide that the Government reserves the right to waive the requirement for bid sample inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. The Government will use the results of the bid sample inspection to determine if contract award is appropriate.

6.4 First article inspection. First article inspection is performed by the manufacturer after contract award. When first article inspection is required (see 6.2), the equipment should be first production units. The first article sample consists of five units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Conformance inspection follows first article inspection. First article inspection is typically witnessed by the Government or a Government representative. The Government reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to ensure compliance with first article inspection.

6.5 Conformance inspection. Conformance inspection is performed by the manufacturer following first article inspection. Conformance inspection is typically witnessed by the Government or a Government representative. The Government reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to ensure compliance with conformance inspection.

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6.6 Definitions.

6.6.1 Accessory. An assembly, a group of parts, or a unit that is not always required for the operation of a test set or unit as originally designed but serves to extend the functions or capabilities of the test set; examples include headphones for a radio set supplied with a loudspeaker, a vibrator power unit for use with a set having a built-in power supply, or a remote control unit for use with a set having integral controls.

6.6.2 Approved equipment test procedure. The test procedure furnished by the Government or furnished by a contractor in accordance with the requirements of the contract or order and approved by the Government in accordance with the contract or order.

6.6.3 Automatic test equipment (ATE). For the purpose of this specification, the terms ATE and automatic checkout equipment (ACE) are synonymous.

6.6.4 Bench-top equipment. Equipment designed to be used on a fixed bench or table or on a mobile cart. Equipment that exceeds 5 kg in weight and has no handles, or that exceeds 20 kg with or without a handle, is considered to be bench-top equipment.

6.6.5 Built-in test (BIT). For the purpose of this specification, the term BIT identifies the test approach using either BITE or self-test capability or both.

6.6.6 Built-in test equipment (BITE). Any device that is part of or permanently mounted in the prime equipment and used for the express purpose of testing that equipment, either independently or in association with external test equipment.

6.6.7 Calibration interval. The maximum length of time between calibration services during which a specified percentage of equipment is expected to remain within acceptable performance levels under normal conditions of handling and use.

6.6.8 Dimensions. Dimensions are specified in units of width, height, and depth of the equipment in its normal operating position, and may be specified as maximum or partial, or both. Maximum dimensions include all handles, latches, knobs, and other protuberances and are the three interior dimensions of a rectangular box into which the equipment (less detachable cords, cables, or accessories) may be fitted. Partial dimensions, when specified, include all protrusions except removable handles, connectors, bumper feet, latches, and cover. Partial depth, when specified, is defined as follows:

a. Portable bench-top equipment. From the front surface of the front panel (less removable bezels, handles, or cover) to all rear protrusions except controls, bumper feet, connectors, latches, or handles.

b. Rack-mounted equipment. From the rack-mounting surface (normally, the rear surface of the front panel) to the rearmost permanent protrusion, including any connectors or controls, but not including cable retractors, slide-out track parts not fixed to the equipment, or cable clearances.

6.6.9 Enclosure. Enclosure and case are interchangeable terms that may apply to any of the classes. The enclosure provides protection for the contained system components during storage, handling, and use and is compatible with the contained class of equipment.

6.6.10 Failure. Equipment failure, as used herein, is any departure from the required performance or operation outside of the required accuracies (not correctable by normal use of the operating controls) or deviation from the criteria of 4.5 after the test is initiated.

6.6.11 Handheld equipment. Any equipment that uses a battery as its primary source of power.

6.6.12 Input power. Input power is defined as follows:

a. Waveform deviation factor. The deviation factor of a wave is the ratio of the maximum difference between corresponding ordinates of the wave and of the equivalent sine wave to the maximum ordinate of the equivalent sine wave when the waves are superimposed in such a way as to make this maximum difference as small as possible. The deviation factor computed from the actual crest factor may exceed the specified deviation factor at peak of the waveform.

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b. Harmonics. Harmonics are defined as follows:

(1) Total harmonics. The total harmonic content is the total rms voltage remaining when the fundamental component is removed and is equal to the square root of the sum of the squares of the voltages of the individual harmonic frequencies.

(2) Individual harmonics. An individual harmonic is the total rms voltage remaining when the fundamental and other harmonic components are removed.

c. Crest factor. The ratio of the crest voltage to the effective voltage that has a value of 1.414 for a true sine wave. The crest factor specified is derived from a total harmonic content (10 percent) wherein the phase and amplitude relationship of the individual harmonic frequencies causes the crest factor to fall between the specified values.

d. Steady-state variation. The effect of periodic and random deviations of instantaneous voltage and frequency of the mean steady-state value, at any constant load, the limit of such effect being expressed as a plus and minus percent of nominal value.

e. Maximum transient amplitude. The maximum excursion of voltage or frequency transient from any initial point within the steady tolerance limits, expressed as a plus and minus percent of nominal voltage or frequency.

f. Transient recovery time. The total elapsed time of the transient disturbance measured from the instant the voltage or frequency (or both) departs from the defined steady-state variation limits to the instant the voltage or frequency (or both) recovers to steady-state condition and stabilizes within the defined steady-state variation band.

6.6.13 Intermittent operation. Alternating periods of operation for the specified time followed by 15 minutes of non-operation.

6.6.14 Lot. The size of a lot is the quantity of units from each individual production run that is specifically set aside for delivery to the Government (i.e., if a given production run consists of 1,000 units of which 150 are set aside for delivery to the Government, the lot size would be 150). Portions of separate production runs will not be combined into a single lot but will be treated as individual lots.

6.6.15 Maintenance and service manual. This manual provides guidance to perform certain levels of maintenance. This information includes parts lists to the component level; schematics and component layout drawings; block and schematic diagrams; a list of required test equipment and connection diagrams; sequential instructions for disassembly, repair, replacement, and reassembly; board-level maintenance and troubleshooting information; and step-by-step instructions for troubleshooting and fault isolation.

6.6.16 Malfunction. An internal failure of the test equipment that occurs during use within its designed parameters and specification that prevents it from further functioning or operating. Examples include (but are not limited to) display lockup, self-test failure, unexplainable error codes, or sudden de-energizing.

6.6.17 Nonrepairable subassembly. A component, module, or subassembly is nonrepairable if the physical nature of the item is such that the item cannot be economically or feasibly repaired due to the excessive cost of material and labor required to effect such repair. This excessive cost is normally considered as 65 percent, or greater, of the acquisition cost of the item. Examples of nonrepairable assemblies are: (1) an integrated circuit, (2) a printed circuit card wherein the components and card are sealed in a hard thermosetting plastic compound, and (3) a module wherein all printed circuit cards or components are sealed in a hard thermosetting plastic compound or the module is hermetically sealed.

6.6.18 Passive device. Equipment that does not analyze, source, or measure a parameter.

6.6.19 Performance requirements of the test equipment. The satisfactory performance of all electrical and mechanical characteristics under the conditioning, destructive, and accelerated tests specified by class and in the acquisition documents for simulating anticipated field service demands as closely as possible.

6.6.19.1 Performance tests, level A and level B. Level A and level B performance tests are characterized by the degree of testing performed for a specific parameter. Level A testing is intended to be a reduced amount of testing per unit that is performed on each unit of a lot. Level B testing is more extensive within each parameter, but is limited to certain samples of the lot.

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6.6.20 Portable equipment. Equipment designed to be easily carried between locations of use. Equipment of 5 kg or less in weight that can be safely moved without handles, or 20 kg or less and provided with a handle, is considered to be portable. Portable equipment may be designed for hand-held (internal power – battery operation) or bench top (external power – commonly AC line voltage) configurations.

6.6.21 Procuring activity. The Military or Federal agency contracting for equipment.

6.6.22 Ruggedized. Physical and operational characteristics that allow equipment to withstand rough handling and unprotected and uncontrolled environments.

6.6.23 Satisfactory operation check (SOC). An SOC employs a maximum number of active circuits of the equipment with a minimum requirement for external input or evaluation equipment to ensure that all functions of the equipment are operational. An SOC is a nominal subset (<10 percent) of the group B performance tests and is used to verify that the equipment is operating within the specified bounds. An SOC is specified by the Government for in-process tests performed during group C and D tests.

6.6.24 Satisfactory operation test (SOT). An SOT employs the necessary external input and evaluation equipment to determine whether the equipment conforms to requirements. An SOT is a subset (<50 percent) of the group B performance tests and is used to verify that the equipment is operating within the specified bounds, usually following group C and D tests.

6.6.25 Self-test. A test or series of tests performed by a device upon itself that shows whether or not the device is operating within designed limits. This includes test programs on computers and ATE that check out their performance status and readiness.

6.6.26 Special tools. Tools not listed in the GSA Global Supply Catalog.

6.6.27 Standby. Equipment is considered to be in a standby condition when any part of its circuitry is energized regardless of the power switch setting.

6.6.28 Test equipment/unit. Test equipment is used to measure, calibrate, gage, test, inspect, diagnose, or otherwise examine materials, supplies, and equipment to determine compliance with specifications, engineering drawings, technical orders, technical manuals, maintenance instructions, or serviceability standards. Test equipment is an instrument with the parts, accessories, components, or any combination thereof, required for performing a specified operational function. Unless otherwise specified (see 6.2), the equipment consists of a single instrument or a mainframe and plug-in configuration. Certain equipment may be complete within it and may not require the addition of parts, accessories, or components to perform a specified operational function. The term “units” is used to denote multiple items of equipment.

6.6.28.1 Mainframe. An integrated system that is composed of interacting or interdependent components and is described by its structure and purpose. A mainframe typically consists of the following: (1) display/readout module, (2) power supply/generation, (3) multiple integration functional areas (electronic circuits), and (4) interface, input/output (I/O) modules, connectors and chassis.

6.6.28.2 Plug-in. A module for a device that can be added to a mainframe/system to provide additional/extra functions or features.

6.6.29 Test plan. A document that provides a listing of tests to be conducted on a given system or item to be tested, the schedule for the tests, the test organizations or personnel involved with the testing, details on the equipment to be tested (configuration and quantities), and disposition of the equipment after each test. A test plan is not normally utilized for only a single test since the test procedure for a single test can contain the necessary information for that test.

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6.6.30 Test procedure. A document that provides detailed step-by-step instructions for conducting a test and includes the identification of prerequisites, test equipment to be utilized for the test, operation and configuration of the equipment to be tested, pass/fail criteria, data to be recorded, and instructions for analyzing data. The test procedure includes blank spaces as necessary for recording data and signatures of any test witnesses. The test witness signature indicates only that the test was conducted in accordance with the test procedure and that the data was accurately recorded; the test witness signature does not indicate approval of the test results or the acceptability of the equipment which was tested. A test procedure should be self-contained and not rely on or refer to other documents, such as the Use and Installation Manual, for conducting the test.

6.6.31 Test report. A document that provides the results of a test and normally consists of a completed test procedure including raw data, analyzed data, test witness signatures, problems that occurred during testing, and actual test equipment utilized for the test.

6.6.32 Use and installation manual. This manual provides a basic overview on the operation of the equipment, to include functions and measurements. It also contains basic error messages and troubleshooting guides.

6.6.33 Ordinary tools. Tools are tools that are listed in the GSA Global Supply Catalog.

6.7 Subject term (key word) listing.

Acoustic noise
ATE
Automatic test equipment
Batteries
BIT
Bounce, loose cargo
Built-in test
Calibration interval
Commercial-off-the-shelf
COTS
Enclosure
Environmental requirements and tests
Explosive atmosphere requirements and tests
Flammable material
Frequency-transient test
Interchangeability
Lithium
Maintainability
Mercury
Radioactive material
Reliability
Shock
Transient-state power
Vibration

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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BID SAMPLE INSPECTION

A.1 SCOPE

A.1.1 Scope. This Appendix provides details regarding bid sample inspection. Bid sample inspection is executed by the Government. This appendix is a mandatory part of the standard when bid sample inspection is required (see 6.2). The information contained herein is intended for compliance.

A.2 BID SAMPLE INSPECTION

A.2.1 Bid sample testing. When a bid sample is required (see 3.1.1 and 6.2), the procuring activity will execute bid sample testing. Unless otherwise specified (see 6.2), bid sample inspection will be performed on two identical units of each model. If readjustment of the equipment is anticipated at the conclusion of any environmental test and prior to the resumption of environmental testing, the rationale for readjustment will identify the environmental tests that should be followed by such readjustment. Bid sample inspection will be performed on the units as specified in 4.5 and as follows:

a. Failure criteria. Failure of any group A through group E test will be considered a major defect. Failure of group F tests will result in elimination from further consideration for procurement.

b. Test sequence. The test sequence will be as specified in A.2.3 through A.2.8 for groups A through F. If group B testing is not required, the test sequence will proceed from group A to groups C and D to group E. Group D testing may be performed concurrently with, or sequentially after, group C testing.

c. Test conditions and control settings. The test conditions and control settings will be in accordance with 4.5.1.

A.2.2 Bid sample process. Bid sample inspection will be in accordance with the steps below and A.2.3 through A.2.8:

a. Submissions. Each manufacturer shall submit two identical units for each model that the manufacturer proposes to have considered for the solicitation. Subject to Government approval, the manufacturer may propose that only one unit be submitted for bid sample inspection; appropriate justification shall be provided by the manufacturer.

b. Tests performed. The Government will perform the bid sample evaluation in accordance with the tests specified in 4.5.

c. Failures. Bid sample testing is anticipated to require 90 business days to complete. To achieve completion in 90 business days, all failures shall be addressed expeditiously. Failures are defined in 4.5.2.5. The testing activity for bid sample testing will notify the sponsor organization of any failure. The manufacturer will then be officially notified of the failure by the Contracting Officer. The manufacturer shall respond within 5 business days. Remedy to the failure shall be accomplished on site at the testing activity location within 10 business days of the official notification. If remedy cannot be accomplished within 10 business days, the manufacturer shall request an extension from the Contracting Officer. Failure to remedy the failure shall result in rejection of the bid sample and elimination of the model from consideration on the solicitation.

d. Elimination. For purposes of bid sample testing, the units submitted for testing are considered a lot. The equipment lot may present only two failures during the bid sample testing. A third failure of the lot shall result in elimination of the model from further consideration on the solicitation. (Elimination after the third failure is a result of extrapolation of the reliability criteria to tangible selection criteria that facilitates conformance to a 90-day bid sample test period and timely execution of the contractual process.)

e. Equipment disposition. All equipment submitted for bid sample testing shall remain with the testing activity until the contract is awarded and the protest period has passed. Return shipping instructions shall be as specified (see 6.2).

A.2.3 Group A. The two units will be subjected to the group A inspection specified in 4.5.3.

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A.2.4 Group B. The two units that have passed group A inspection (see A.2.3) will be subjected to the group B tests of 4.5.4. If group B testing is not required, the test sequence will proceed from group A to groups C and D, as applicable.

A.2.5 Group C. One unit that has passed group B testing (see A.2.4) will be subjected to the group C tests of 4.5.5.

A.2.6 Group D. One unit that has passed group C testing (see A.2.5) will be subjected to the group D tests of 4.5.6.

A.2.7 Group E. The unit that has passed group C testing (see A.2.5) and group D testing (see A.2.6) will be subjected to the group E tests of 4.5.7. The number of units tested will be as follows:

a. Maintainability. One unit will be tested.

b. Acoustic noise. One unit will be tested. Equipment that does not contain a motor, such as a fan or blower, or other audible noise source will not be tested.

c. Dimensions and weight. Two units will be tested.

d. Mechanical stability. Two units will be tested. Units smaller than 2000 cc with a weight less than 2.5 kg will not be tested.

e. Equipment emanation. Two units will be tested. X-radiation tests will not be performed unless the equipment is designed to have operating potentials of 10 kV or greater or could have potentials of 10 kV or greater due to the failure of any one component.

A.2.8 Group F. Group F tests will be performed by evaluating the manufacturer’s data to establish that reliability falls within the parameters outlined in 3.14.

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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 84

Review activities:
Army – AR, AV, CE, EA, MD, MI, TE
Navy – AS, MC, OS, YD
Air Force – 13, 19

Preparing activity:
Navy – SH
(Project 6625-2019-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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